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SCHOLARLY REVIEW

# Navigating the Risks Beyond the Label: Unpacking Global Nutritional Supplement Safety

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Nutritional supplement use is common among athletes aiming to enhance performance, recovery, and health. However, variable regulatory frameworks and limited safety oversight create risks for inadvertent doping violations. This article provides a global overview of supplement use, relevant authorities, legislation, and safety measures, with a focus on third-party testing (TPT) as a risk-mitigation strategy. Data from six global regions—Africa, Asia, Australia/New Zealand, Europe, Latin America, and North America—were synthesized from peer-reviewed studies, governmental sources, and regional expert contributions. Reported supplement use ranged from 7% to 100% among athletes (variability within regions), with protein powders, vitamins/minerals, creatine, caffeine, and sports drinks being most prevalent. High-risk products (potential anti-doping rule violations), including certain herbal blends, preworkouts, and weight-management supplements, were reported across all regions. While some countries have robust regulatory systems, most lack harmonized or enforceable safety frameworks. TPT programs, which independently verify products for prohibited substances, remain concentrated in the global northwest (Europe, North America, and Australia/New Zealand); awareness and use of TPT certification vary widely, and even in regions with established systems, athlete adherence is inconsistent. Barriers to low-risk supplement use are limited TPT availability, cost, differences in labeling (including language), and cultural factors. Firsthand experiences and perceptions highlight widespread misconceptions about supplement safety and certification. The authors recommend expanded athlete and team-around-the-athlete education, improved global access to TPT low-risk supplements, and policy initiatives to harmonize safety standards. This work emphasizes the need for coordinated international efforts to protect athlete health and integrity while allowing access to evidence-based supplementation.

***Keywords:*** contamination, doping, adulteration, spiking, third-party testing

Athletes across the world commonly use nutritional supplements (Knapik et al., 2016). In sports, nutritional supplements often constitute a mixture of dietary supplements, sports foods, and functional foods with ergogenic claims (Wardenaar et al., 2017), with higher prevalence at the elite level (Knapik et al., 2016). As many as 100% of athletes report the use of nutritional supplements during their career (Daher et al., 2022), with reasons for use including to promote performance, recovery,

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and/or general health (Vento & Wardenaar, 2020; Wardenaar et al., 2017).

There is no single global definition, either legal or within nutritional science, of what constitutes a dietary supplement (Maughan, Burke, et al., 2018). The consensus statement on dietary supplements by the International Olympic Committee describes supplements as a food, food component, nutrient, or non-food compound that is purposefully ingested in addition to the habitually consumed diet with the aim of achieving a specific health and/or performance benefit (Maughan, Burke, et al., 2018). In this work, the term "nutritional supplement" is used as an umbrella term for dietary supplements, sports foods, and ergogenic supplements. Although there is insufficient evidence to support the effectiveness of many supplements, there is growing recognition that some supplements support athlete health and performance beyond normal dietary food intake (Backhouse, 2023). Proposed reasons include, but are not limited to: (a) the assertion that some nutrients may be difficult to obtain in sufficient quantities from food (e.g., creatine, beta-alanine, vitamin D/B12 and iron); (b) athletes may exclude foods rich in specific nutrients (e.g., omega-3 fatty acids from fatty fish, or iron from red meat) due to individual food preferences or adherence to dietary patterns such as vegetarianism or veganism, or extreme weight cutting; (c) exact doses of the substance are difficult to provide with normal foods (e.g., caffeine); (d) certain supplements may be a more convenient source of specific nutrients during exercise (e.g., carbohydrate sports foods as part of competition); and (e) supplements can be an alternative to food in situations when food hygiene (e.g., during times of traveling for training or competition) or food contamination (e.g., use of clenbuterol in farming) is of concern (Close et al., 2022).

Online shopping, social media, and globalization have increased the awareness, exposure, and access to nutritional supplements worldwide. Although countries and regions may have some harmonization to regulate their respective nutritional supplement markets, no clear legislation toward supplement and/or ingredient safety is available worldwide (Tallon & Kalman, 2025). At the same time, safety regulations on a national level often lack well-developed frameworks (Ng et al., 2022). Despite (regional) legislation, the consumption of nutritional supplements and certain foods may come with risk, such as the consumption of heavy metals (Ćwieląg-Drabek et al., 2020). However, specifically for elite athletes, supplement use may be associated with the risk of anti-doping rule violations (Lauritzen, 2022; Maughan, Burke, et al., 2018), based on World Anti-Doping Agency (WADA) regulations (World Anti-Doping Agency, 2021). Analytical studies have shown that 14%–50% of tested nutritional supplements tested positive for prohibited substances (Jagim et al., 2023). When adhering to the WADA code, doping rules are uniform across the world. However, there are sport associations with their own regulations, such as the National Collegiate Athletic Association in the United States. The WADA was established in 2004 and oversees anti-doping worldwide. Still, there are likely major differences in the available resources to educate athletes on safe and low-risk supplement use and the availability of risk-mitigating strategies toward the use of safer (i.e., low risk), evidence-based nutritional supplements across regions. Education is recognized as one factor available to influence athlete supplementation behaviors. However, behavioral change involves complex psychological, social, and cultural dimensions that require distinct methodologies and theoretical frameworks. Addressing these aspects adequately would go beyond the objectives and constraints of the present work, which is centered on infrastructural and systemic factors influencing supplement safety and low-risk use.

To minimize the risk of anti-doping rule violations while ensuring the effective use of nutritional supplements, athletes should prioritize products that have the following characteristics: (a) supported by substantial scientific evidence, (b) manufactured according to good clinical practice, (c) cross-checked for ingredient profiles, and (d) verified by independent third-party testing (TPT) for prohibited substances in sport (Jagim et al., 2023). Education and monitoring initiatives to promote low-risk supplement use are available in some countries. However, the majority still lack effective systems for monitoring, quality assurance, and supplement control (Djaoudene et al., 2023), along with limited access to TPT options. For athletes in particular, the use of TPT supplements, those tested for substances banned in sport by independent processes and organizations, is vitally important, as TPT currently represents the only reliable method to determine whether a product poses a substantial risk of containing WADA-banned substances (Eichner et al., 2019). Even in countries where TPT systems are widely available, there may be low awareness of, or inconsistent adherence to their use (Gimbar et al., 2025; McDaid et al., 2024; Schott, Seltzer, et al., 2024; Vento & Wardenaar, 2020; Wardenaar et al., 2020).

Although National Olympic Committees (NOCs) and National Anti-Doping Organizations (NADOs) typically follow standards set by WADA, factors such as an athlete's caliber and cultural norms within sport may influence supplement use and adherence to these safeguards (Henning & Dimeo, 2014). Language barriers may also limit access to TPT systems developed in other regions, while physical or political isolation may further restrict availability. Finally, financial differences between countries may reduce access to nutritional supplements (Bayati et al., 2025), particularly the more costly TPT-verified supplements, limiting opportunities for some athletes to use them.

Nutritional supplement use carries inherent risks, and its prevalence among high-performance athletes varies substantially across regions. This global heterogeneity is shaped by cultural norms, levels of access, regulatory frameworks, and educational resources. As a result, athletes, coaches, and support personnel may struggle to contextualize supplement use practices in other countries, particularly relevant during times of international travel for training and competition. Assumptions about supplement quality, safety, or prevalence can be misleading—potentially influencing international competition preparedness. These complexities highlight the need for globally informed education strategies and policies in elite sport. This will allow all athletes around the globe to benefit from (the potential) beneficial effects of nutritional supplements that are low risk to use without risking a positive doping test. This article offers a global regional overview of supplement use, legislation, and risk-mitigation strategies aimed at supporting low-risk supplement use and reducing the likelihood of anti-doping rule violations. Additionally, this article includes anecdotes from the field describing current limitations, briefly discusses the outcomes, and lists suggestions for future optimization of athlete safety regarding low-risk nutritional supplement use.

# Methods

## Regional Classification

Six global regions were defined to reflect the current state of nutritional supplement use: (a) Africa, (b) Asia, (c) Australia and New Zealand, (d) Europe (including Russia), (e) Latin America (including Mexico), and (f) North America. In most cases, the regions align with continental boundaries; however, Mexico was grouped with Latin America due to shared language (Spanish or



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Portuguese) and stronger cultural ties with that region than with the United States and Canada.

## Author Contributions and Regional Representation

Each region was represented by two to three co-authors. These individuals had expertise in one or more of the following areas: performance nutrition, (ergogenic) nutritional supplement research, supplement safety and/or anti-doping, or nutritional supplement industry practices. In cases where no region-wide legislation could be identified, representatives described the situation in the country where they lived or worked. This approach provided a range of insights into regional diversity by combining desk research with input from local performance nutritionists. However, the overview is not intended as a comprehensive account of all national regulations and practices.

## Data Extraction and Content Description

### *Athlete Supplement Behavior*

For this article, nutritional supplement use includes dietary supplements, sports foods, and supplements with ergogenic claims (Wardenaar et al., 2017). The focus was on published use among elite adult athletes (McKay et al., 2022), based on studies published between 2014 and 2025. Each group of co-authors representing a region was asked to list relevant studies for their area. They were instructed to use common search engines (such as PubMed and Google Scholar) to identify relevant studies. No predetermined search instructions were provided to each of the regions, but per region, there was at least one co-author who could be considered a content expert familiar with the published literature representing their region. If only a limited number or no studies fitting these criteria were available, other relevant studies published as early as 2000 were included. For each study, the distribution for sex and age was listed. In case studies that investigated junior/adolescent athletes next to adult athletes, this was highlighted. Athlete classification was stated from Tier 1, reflecting recreationally active individuals, up to Tier 5, reflecting world-class athletes (McKay et al., 2022); however, when athlete level/tier was not clearly stated in the original study, categorization was inferred based on the descriptive data provided within the respective studies. Additionally, the investigational time frame when supplements were used by athletes is stated as reported in the original studies: current, regular, last month, during the season, last year, lifetime use, or as undefined when not specified. This time frame is based on the period during which the questionnaire was used to evaluate supplement use.

For each study population, the total prevalence of nutritional supplement use was documented. Where available, the average number of supplements used and the prevalence of the most commonly reported supplements were also included. Products include dietary supplements (e.g., multivitamins, essential fatty acids, or individual vitamins and minerals), sport foods (e.g., sports drinks, recovery products, or other macronutrient-containing items), and supplements with ergogenic claims (e.g., caffeine, creatine, beta-alanine, nitrate, and sodium bicarbonate). The reporting of prevalent supplements depended on the level of detail provided in the original studies.

Additionally, supplements considered higher risk for contamination—and thus more likely to lead to an anti-doping rule violation—were identified based on common ergogenic claims. These include products marketed as supplementals reducing fat mass (e.g., fat burners such as ephedra; Batsis et al., 2021; Cohen et al., 2018, 2021); boost testosterone, stimulate muscle growth, or act as stimulants (e.g., Tribulus terrestris, prohormones, ashwagandha; Abbate et al., 2015; Barker et al., 2025; Eichner & Tygart, 2016; Geyer et al., 2004); contain botanicals or herbs (e.g., ginseng, Ginkgo biloba; Qin et al., 2021); or serve as preworkout blends, containing multiple stimulating ingredients and are frequently not well defined (Lauritzen, 2022). Finally, supplements containing amino acids (e.g., branched-chain amino acids, weight gainers) have been listed as supplements with a higher risk for contamination (Alaedini et al., 2021; Lee et al., 2020; Micalizzi et al., 2021). Although whole protein supplements have historically also been subject to contamination (Alaedini et al., 2021; Geyer et al., 2004), they were not listed separately in Table 1 as they are commonly classified within the most frequently used supplement category. While reporting supplement use by region, the original terminology used in each source (e.g., dietary supplement vs. nutritional supplements) was retained in the results tables. In text, "nutritional supplement" is used as an umbrella term for dietary supplements, sports foods, and ergogenic supplements when not linked to a specific study context or regulation.

### *Authorities and Legislation*

The analysis assessed whether nutritional supplement harmonization—the alignment of laws, regulations, standards, and practices across nations to create a unified policy framework for supplement safety—was present within each region. In addition, at least three country-specific examples were identified to illustrate national authorities (i.e., an organization or body officially designated by a government to regulate, oversee, or enforce rules related to nutritional supplements), and legislation focused on regulating supplements and/or sports foods. In most cases, these country-specific examples are related to the country of origin of the co-authors.

### *Education and Risk Mitigation Systems*

A brief review was undertaken on available TPT organizations per region (broad internet search), and if systems from other regions were available. Second, education efforts per region were listed. While potentially depicting a (selection and/or availability) biased approach, this search focused on national governments, NOCs, NADOs, and Regional Anti-Doping Organizations (RADOs). However, these organizations are often considered reliable sources of information, and efforts were made to provide a (nonexhaustive) overview. Additionally, a brief supplementary review was conducted to identify other potentially relevant organizations engaged in educational initiatives related to nutritional supplements, using a broad internet search. Third, limitations and barriers to accessing low-risk nutritional supplements were identified, which were mainly based on the availability, or lack of availability, of TPT supplements. When available, examples of efforts to address these barriers were also included. In some cases, ethical and policy discussions around accessibility, affordability, and efficacy were noted as part of these regional considerations.

### *First-Hand Experiences and Perceptions on Awareness of Low-Risk Supplement Use*

The manuscript includes relevant anecdotes from conversations that the authors had with stakeholders that highlight gaps in awareness of TPT. The individual contributed anecdotes derived from the memory of the authors, which, while subjective, offered contextually relevant insights. These examples illustrate confusion, lack of understanding, or inconsistent knowledge about the need for testing and the credibility of TPT organizations. No names are



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**Table 1 Most Frequently Reported Supplements and Risky Supplements by Elite Athletes Per Region**

| Study | Country | Athlete population | Time frame | Nutritional supplement use (most frequent) | Supplements at risk[a] |
|---|---|---|---|---|---|
| Africa | | | | | |
| Nyawose et al. (2022) | South Africa | $n = 343$ (100% male, 0% female), semiprofessional soccer players<br>Age: 24 ± 4 years<br>Tiers 2–3 | Last month | Supplement use: 41%<br>Supplements used: sports beverages, 23%; energy beverages, 6%; dietary supplements, 5%; protein shakes, 3%<br>Types of dietary supplements were not further specified | |
| Mse et al. (2021) | Kenya | $n = 239$ (57% male, 43% female), elite runners<br>Age: Range from under 18 to above 35<br>Tiers 3–5 | Lifetime use | Supplement use: 41%<br>The most consumed supplements (numbers undefined): carbohydrates, proteins, and vitamins | Traditional herbs, 7% |
| Kisten and Naidoo (2019) | South Africa | $n = 449$ (100% male, 0% female), development soccer clubs<br>Age: Undefined, but this included multiple first under-19 teams<br>Tier 2 | Lifetime use | 25% reported currently using supplements, 33% have used supplements in the past, and 42% never used supplements | |
| Tawfik et al. (2016) | Egypt | $n = 358$ (56% male, 44% female), including 76 competitive athletes, mainly team sports<br>Age: 14 ± 2 years (range 13–18)<br>Tiers 1–3 | Undefined | Supplement use: 49%<br>Supplements used: sport drinks, 33%; creatine, 24%; vitamins and minerals, 26%; amino acids, 22% | Amino acids, 22% |
| Mkumbuzi (2015) | Zimbabwe | $n = 86$ (100% male, 0% female), professional soccer players<br>Age 23 ± 2 years<br>Tier 3 | During the season | Supplement use: 56% (of which 21% daily)<br>Supplements used: vitamin supplements, 20%; mineral supplements, 17%; protein powders, 9%; herbal supplements, 4% | Herbal supplements, 4% |
| Asia | | | | | |
| Shahzad et al. (2024) | Pakistan | $n = 100$ (sex-ratio undefined), national or international professional weightlifters<br>Age: 33 years (mean)<br>Tiers 3–4 | Current use | Supplement use: 100%<br>Supplements used: multivitamin, 84%; vitamin D, 84%; vitamin C, 80%; vitamin E, 76%; sports drinks, 65%; vitamin B, 65%; amino acids, 62%; creatine, 51% | Ginseng, 33%; Ginkgo biloba, 29%; Weight gainer, 27%; Ephedra, 18%; slimming products, 15% |
| Deota and Chandorkar (2024) | India | $n = 81$ (67% male, 33% female), elite cricketers from the following teams: boys ($n = 26$), Under-23 boys ($n = 28$), senior women ($n = 12$), and Under-19 women ($n = 15$)<br>Age: undefined<br>Tiers 2–4 | Regular use | Supplement use: 100%<br>Supplements used: sports drinks, 96%; protein, 44%; BCAA, 3% | BCAA, 3% |
| Thapa et al. (2023) | Nepal | $n = 270$ (50% male, 50% female), national athletes, mainly team sports (79%)<br>Age: 26 ± 4 years<br>Tier 3 | Current use | Supplement use: 7%<br>Supplements used by the ones that reported the use: whey protein, 75%; multivitamin, 25% | |
| Meiyao et al. (2021) | China | $n = 295$ (0% male, 100% female), sports university students<br>Age: undefined<br>Tiers 1–2 | Current use | Supplement use: 85%<br>Supplements used: undefined | |

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**Table 1** *(continued)*

| Study | Country | Athlete population | Time frame | Nutritional supplement use (most frequent) | Supplements at risk[a] |
|---|---|---|---|---|---|
| Prabowo et al. (2022) | Indonesia | $n = 125$ (93% male, 7% female), elite amateur boxers<br>Age: 24 years (mean)<br>Tier 3 | Regular use | Supplement use: 100%<br>Supplements used: multivitamin/mineral, 87%; fat burner, 31%; BCAA, 21%; whey protein, 14%; creatine, 12% | Fat burner, 31%; BCAA, 21% |
| Sadek et al. (2022) | Lebanon | $n = 455$ (73% male, 27% female), athletes from Lebanese sports federations (including 55% amateur athletes)<br>Age: 28 ± 8 years<br>Tier 3 | Current use | Supplement use: 74%<br>Supplements used: vitamin and mineral supplements, 36% | |
| Rashani et al. (2021) | Sri Lanka | $n = 386$ (67% male, 33% female), national athletes attending the 2018 Asian Games<br>Age: 26 ± 5 years<br>Tiers 3–4 | Last year | Supplement use: 92%<br>Supplements used: multivitamin, 57%; electrolytes, 50%; protein, 44%; calcium, 20%; creatine, 17% | Herbal and botanicals, 14%; weight gainers, 1% |
| Tabata et al. (2020) | Japan | $n = 299$ (54% male, 46% female), elite track-and-field athletes (subsample from a study with $n = 574$ athletes, including 275 junior athletes)<br>Age: 25 ± 4 years<br>Tiers 3–4 | Current use | Supplement use: 69%<br>Supplements used: vitamins, 54%; amino acids, 54%; minerals, 29%; protein, 20%; creatine, 12% | Amino acids, 54%; herbal supplements, 7% |
| Aljaloud (2018) | Saudi Arabia | $n = 164$ (58% male, 42% female), athletes participating in the 2016 Rio Olympic Games<br>Age: 20 years and older<br>Tier 4 | Current use | Supplement use: 71%<br>Supplements used: sports drinks, 16%; protein, 15%; iron, 14%; herbal supplements, 13% | Herbal supplements, 13% |
| Balaravi et al. (2017) | Malaysia | $n = 50$ (66% male, 34% female), national elite athletes<br>Age: 22 years (IQR = 6)<br>Tiers 3–4 | Current use | Supplement use: 74%<br>Supplements used: energy/sports drinks, 84%; protein drinks/powder, 54%; multivitamin, 54% | Ginkgo biloba, 24%; Ginseng, 16%; Other traditional/herbal supplements, 18% |
| Ling et al. (2017) | Malaysia | $n = 77$ (74% male, 26% female), team sports athletes of the National Sports Institute<br>Age: 23 ± 4 years<br>Tiers 2–3 | Current and past use | Supplement use: 60% with an average of 1.5 supplements<br>Supplements used: sports foods/drinks, 46%; protein, 25%; vitamins and minerals, 27%; health foods, 12%; other nutritional ergogenic aids, 14% | Weight gainer, 4% |
| Seifbarghi et al. (2015) | Iran | $n = 234$ (100% male, 0% female), professional soccer players<br>Age: undefined<br>Tier 3 | During season | Supplement use: 100%<br>Supplements used: vitamin C, 62%; vitamin E, 54%; magnesium, 39%; multivitamin, 33% | |
| Sato et al. (2015) | Japan | $n = 552$ (47% male, 53% female), athletes who participated or prepared for the 2012 London Olympic Games<br>Age: 25 ± 5 years<br>Tiers 3–4 | Last year | Supplement use: 82%<br>Supplements used: amino acids, 56%; protein, 21%; vitamins, 20%; minerals, 15% | Amino acids, 56%; oriental herbs, 1% |
| Amirsasan (2014) | Iran | $n = 42$ (86% male, 14% female), athletes participating in the 2012 London Olympic Games<br>Age: 25 ± 3 years<br>Tier 4 | Lifetime use | Supplement use: 93%<br>Supplements used: vitamin B-complex, 92%; vitamin C, 83%; glutamine, 81%; multivitamin, 81%; whey protein, 57%; amino acids, 55% | Glutamine, 81%; amino acids, 55% BCAA, 41% |

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**Table 1** *(continued)*

| Study | Country | Athlete population | Time frame | Nutritional supplement use (most frequent) | Supplements at risk[a] |
|---|---|---|---|---|---|
| Golshanraz et al. (2014) | Iran | $n = 350$ (sex-ratio undefined), national-level athletes<br>Age: 22 years (mean)<br>Tier 3 | Current use | Supplement use: percentage not defined<br>Supplements used for male and female, respectively: vitamins, 76.9%/77.4%; energy supplements, 56.9%/47.1%; minerals, 44.6%/57.4% | |
| Aljaloud and Ibrahim (2013) | Saudi Arabia | $n = 105$ (100% male, 0% female), professional soccer players<br>Age: 26 ± 3 years<br>Tier 3 | During the season | Supplement use: 93%<br>Supplements used: Sports drinks, 89%; Vitamin C, 83%; Calcium, 68%; Health bars, 59%; Multivitamin, 52% | Slimming products, 29%; Ginseng, 29%; amino acids, 27%; weight gainer, 22%; Ephedra, 13%; Ginkgo biloba, 10% |
| Australia/New Zealand | | | | | |
| Sport Integrity Australia 2025 (Positive Behaviours in Sport Study, n.d.; in preparation; Study website) | Australia | $n = 164$ (59% male, 41% female in total sample), athletes at national or international level (subsample from a study with $n = 879$, including all levels of sports)<br>Ages: 15 and older<br>Tiers 3–4 | Last year | Supplement use: 40%<br>52% of national/international athletes reported the use of a nutritional supplement<br>The type of supplement used was not recorded | |
| Waller et al. (2019) | Australia | $n = 94$ (41% male, 59% female), at least national-level athletes on sports scholarships<br>Age: 20 ± 5 years (range 12–38)<br>Tiers 3–4 | Last year | Supplement use: 87% (average 4 ± 4 supplements)<br>Supplements used: sports drinks, 70%; caffeine, 48%; protein powder, 42%; sports bars, 42%; vitamin C, 33%; meal replacement drinks, 29% | Exotic berries, 12% |
| Shaw et al. (2016b) | Australia | $n = 39$ (49% male, 51% female), elite and world-class swimmers<br>Age: 22 ± 3 years<br>Tiers 4–5 | Last year | Supplement use: 97%<br>Supplements used: sports drinks, 92%; vitamin C, 85%; zinc, 85%; sports bars, 64%; liquid meal supplements, 64%; caffeine, 49%; protein powders, 46%; multivitamin, 46% | Echinacea, 15%; fat burner, 8%; Tribulus terrestris, 3%; amino acids, 3% |
| Dascombe et al. (2010) | Australia | $n = 72$ (50% male, 50% female), at least national-level athletes participating in kayaking, field hockey, rowing, water polo, swimming, athletics, and netball<br>Age: 22 ± 4 years<br>Tiers 3–4 | Current use | Supplement use: 87% (this averaged 1.5–6.0 supplements per athlete)<br>Supplements used: minerals, 46%; vitamins, 43%; iron, 31%; caffeine, 22%; protein, 17% | |
| Hellemans et al. (2009) (a report to Drug Free Sport NZ—in preparation) | New Zealand | $n = 398$ (52% male, 48% female), elite or developing athletes from 21 different sports<br>Age: 25 ± 7 years<br>Tiers 2–4 | Last 6 months | Supplement use: 93%<br>Median supplement use was 3 (range, 1–14)<br>Supplements used: carbohydrate and energy supplements, 53%; protein and amino acid supplement, 47%; vitamin and minerals, 47%; sport and energy bars, 16%; creatine, 15% | Protein and amino acid supplement, 47%; herbals, 12%; HMB, 6%; dehydroepiandrosterone, testosterone boosters, Tribulus terestris, 3%; thermogenetics/fat burners, 2% |
| Baylis et al. (2001) | Australia | $n = 77$ (52% male, 48% female), national-level up to world-class swimmers<br>Age: not defined<br>Tiers 3–5 | Last 6 months | Supplement use: 99%, totaling 207 different supplements<br>Supplements used: sports drinks, 87%; vitamin C, 73%; multivitamin/mineral, 58%; Echinacea, 39%; liquid meals, 37%; creatine, 31%; vitamin E, 31% | Echinacea, 39%; protein/amino acids, 18%; Ginseng, 8%; Chinese preparations, 4%; Evening primrose, 3% |

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**Table 1** *(continued)*

| Study | Country | Athlete population | Time frame | Nutritional supplement use (most frequent) | Supplements at risk[a] |
|---|---|---|---|---|---|
| Europe | | | | | |
| Sousa-Rufino et al. (2025) | Spain | $n = 129$ (71% male, 29% female), professional soccer players<br>Age: 27 ± 5 years<br>Tiers 3–4 | Lifetime | Supplement use: 96%<br>Supplements used: creatine, 67%; caffeine, 57%; whey protein, 48%; sports drinks, 37%; vitamin D, 28%; sports bars, 27%; magnesium, 21%; vitamin C, 20% | Melatonin, 12%; BCAA, 12%; curcumin, 11% |
| Isenmann et al. (2024) | Germany | $n = 409$ (47% male, 53% female), national- and international-level athletes<br>Age: 12–18 years (26%), 19–25 years (42%), 26–35 years (27%), >35 years 19 (5%)<br>Tiers 3–4 | Recently used | Supplement use: 70%<br>The highest prevalence was reported in power sports such as powerlifting (88%), with the lowest prevalence in technical sports such as gymnastics (59%)<br>Supplements used: undefined | |
| Newbury et al. (2023) | United Kingdom | $n = 11$ (49% male, 51% female), national-level swimmers of 16 years and older (subsample from a study with $n = 44$ athletes, including junior athletes)<br>Age: 20 ± 2 years<br>Tier 3 | Recently used | Supplement use: 98%<br>Supplements used: protein powder, 82%; caffeine, 82%; vitamin D3, 76%; creatine, 55%; omega-3 fatty acids, 55%; protein enhanced foods, 45%; beta-alanine, 45%; zinc, 45% | |
| McDaid et al. (2023) | Northern Ireland and Republic of Ireland | $n = 55$ (51% male, 49% female), international championship level athletes (subsample from a study with $n = 217$, including amateur athletes)<br>Age: 18+<br>Tiers 2–4 | Current use | 60% and 59% of elite and amateur athletes reported nutritional supplement use, respectively<br>Supplement used in elite athletes: protein, 22%; vitamin D, 14%; creatine, 14%; vitamin C, 12%; fish oils, 8%<br>Supplements used in amateur athletes: protein, 18%; vitamin D, 16%; vitamin C, 14%; creatine, 12%; fish oils, 9% | In amateur athletes: Turmeric/curcumin, 14%; Ashwagandha, 10%; CBD, 6%; Black maca, 4% |
| Mata et al. (2021) | Spain | $n = 49$ (37% male, 63% female), fencers at national ($n = 16$) and international ($n = 33$) level<br>Age: 22 ± 6 years<br>Tiers 3–4 | Consumed in the past | Supplement use: 47%. Supplements used: sports drinks, 45%; vitamin C, 43%; sports bars 39%; caffeine, 29%; whey protein, 16%. | Green Tea, 18%; BCAA, 10% |
| Sánchez-Oliver et al. (2020) | Spain | $n = 144$ (58% male, 42% female), rugby players ($n = 69$ professional, $n = 75$ amateur)<br>Age: 24 ± 5 years (professional) and 25 ± 5 years (amateurs)<br>Tiers 2–3. | Regular use | Supplement use: 65%<br>Supplements used: whey protein, 44%; caffeine, 42%; sports drinks, 38%; energy bars, 34%; creatine, 31%; BCAA, 19% | BCAA, 19% |
| Aguilar-Navarro et al. (2021) | Spain | $n = 504$ (65% male, 35% female), elite athletes participating in individual and team sport<br>Age: 15–66 years (range)<br>Tiers 3–4 | Regular use | Supplement use: 65% (male) and 57% (female). On average, they both reported 3 supplements<br>Supplements used, males: protein, 50%; amino acids/BCAA, 38%; multivitamin, 34%; glutamine, 26%; creatine, 24%<br>Supplements used, females: multivitamin, 39%; amino acids/BCAA, 39%; protein, 29%; glutamine, 22%; iron, 22%; recovery supplements. 21% | Males: amino acids/BCAA, 38%; glutamine, 26%<br>Females: amino acids/BCAA, 39%; glutamine, 22% |

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**Table 1** *(continued)*

| Study | Country | Athlete population | Time frame | Nutritional supplement use (most frequent) | Supplements at risk[a] |
|---|---|---|---|---|---|
| Baranauskas et al. (2020) | Lithuania | $n = 247$ (75% male, 25% female), elite athletes training for Olympic sports (team and individual)<br>Age: not defined<br>Tiers 3–4 | Regular use | Supplement use: percentage not defined<br>Supplements used: carbohydrates, 86%; vitamins, 81%; minerals, 75%; protein, 70%; multivitamin, 62%; caffeine, 36%; omega-fatty acids, 47%; creatine, 25% | Herbal supplements, 19% |
| Hurst et al. (2021) | United Kingdom | $n = 583$ (77% male, 23% female), from team (78%) and individual (22%) sports, who competed at club (26%), county (37%), national (28%), and international (9%) level<br>Age: 21 ± 4 years<br>Tiers 2–4 | Current use | Supplement use: 53%<br>A total of 29% used one supplement, 18% reported the use of two supplements, and 6% reported the use of three supplements<br>Supplements used: ergogenic supplements (e.g., amino acids, creatine and sodium bicarbonate), 42%; sport foods and drinks (e.g., sports drinks, sports gels and sports bars), 21%; medical supplements (e.g., calcium, iron, probiotics), 18%; superfoods (e.g., herbs, Goji berries, Maca), 2% | Ergogenic supplements (e.g., amino acids, creatine and sodium bicarbonate), 42%; superfoods (e.g., herbs, Goji berries, Maca), 2% |
| Baltazar-Martins et al. (2019) | Spain | $n = 527$ (66% male, 34% female), high-performance athletes from individual and team sports<br>Age: 15–41 years (range)<br>Tiers 3–4 | Last year | Supplement use: 64% (median three supplements per athlete, with a range of 1–12)<br>Supplements used: protein, 41%; amino acids/BCAA, 37%; multivitamin, ~33%; glutamine, ~20%; creatine, ~17% | Amino acids/BCAA, 37%; glutamine, ~20% |
| Sekulic et al. (2019) | Croatia and Kosovo | $n = 912$ (61% male, 39% female), professional team sport athletes involved in four Olympic sports: Basketball ($n = 228$), soccer ($n = 324$), volleyball ($n = 154$), and handball ($n = 206$)<br>Age: 22 ± 3 years<br>Tiers 3–4 | Regular and irregular use | 13% of athletes reported regular nutritional supplement use, 36% occasionally<br>Supplements used: Vitamin/minerals, 67%; isotonic drinks, 59%; energy bars, 58%; iron, (40%); recovery supplements, 40%; carbohydrates, 37%; protein/amino acids, 36%; creatine, 11% | Protein/amino acids, 36%; "other" supplements (including Tribulus Terrestris), 9% |
| Wardenaar et al. (2017) | The Netherlands | $n = 778$ (52% male, 48% female), elite athletes with a Dutch Olympic Committee athlete status<br>Age: 25 ± 10 years<br>Tiers 3–4 | Lifetime | Supplement use: 97%<br>Supplements used: multivitamin, 63%; isotonic sports drinks, 46%; recovery drinks, 46%; energy bars, 44%; protein shake, 42% | BCAA and amino acids (14/9%); herbal supplements (2/1%) |
| Sousa et al. (2013) | Portugal | $n = 292$ (68% male, 32% female), athletes from Portuguese national teams<br>Age: 20 ± 5 (of which 69% ≥18 years)<br>Tier 3 | Last year | Supplement use: 66% of (median 4 supplements per athlete, with a range of 1–16)<br>Supplements used by users: multivitamin/mineral, 67%; sport drinks, 62%; magnesium, 53%; protein, 50%; glutamine, 33% | Glutamine, 33% |
| Latin America | | | | | |
| Gonzales-Krapp et al. (2024) | Peru | $n = 238$ (47% male, 53% female), elite Peruvian athletes at the 2019 Pan American Games<br>Age: 15–45 years (range), of which 75% in the range of 21–30 years<br>Tiers 3–4 | Undefined | Supplement use: 94%<br>Supplements used: protein, 85%; preworkout, 66%; rehydrating beverages, 56%; iron, 40%; creatine, 37% | Preworkout, 66%; amino acids, 19%; weight gainers, 14%; energizers, 6% |

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**Table 1** *(continued)*

| Study | Country | Athlete population | Time frame | Nutritional supplement use (most frequent) | Supplements at risk[a] |
|---|---|---|---|---|---|
| de Oliveira et al. (2023) | Brazil | $n = 239$ (44% male, 56% female), athletes ranging from regional to international level<br>Age: 27 ± 9<br>Tiers 3–4 | Current use | Supplement use: 63%<br>Supplements used: sport drinks, 56%; energy drinks, 38%; protein, 20%; amino acids, 19% | Amino acids, 19% |
| Henriquez et al. (2021) | Chile | $n = 186$ (76% male, 34% female), disabled athletes from different sports at the national or international level<br>Age: 30 ± 10 years<br>Tiers 3–4 | Current use | Supplement use: 21%<br>Supplements used by users: protein, 87%; vitamins, 44%; creatine, 33%; amino acids, 23%; weight gainers, 3% | Amino acids, 23%; weight gainers, 3% |
| Nabuco et al. (2017) | Brazil | $n = 182$ (83% male, 17% female), athletes participating in regional or international events<br>Age: 24 ± 7 years<br>Tiers 3–4 | Current use | Supplement use: 47% (average 3.0 ± 2.5 supplements per athlete, ranging from 1 to 12)<br>Supplements used: whey protein, 25%; BCAA, 15%; maltodextrin, 14%; glutamine, 9%; creatine, 7%; multivitamin, 7% | BCAA, 15%; glutamine, 9%; thermogenic supplements, 5%; preworkout, 2%; prohormones, 2% |
| North America | | | | | |
| Schott, Seltzer, et al. (2024) | United States | $n = 410$ (47% male, 53% female), NCAA DI collegiate athletes<br>Age: 21 ± 2 years<br>Tier 4 | Last year | Supplement use: 91%<br>Supplements used: sports drinks, 72%, protein shakes, 65%; recovery drinks, 57%; sports bars, 56%; caffeine, 56% | For male and female, respectively: preworkout, 29%/7%; BCAA, 12%/1%; weight gainers, 9%/2%; herbs, 5%/3%; Ashwagandha, 5%/3%; Maca root, 2%/2%; Fenugreek, 2%/2%; HMB, 1%/0% |
| Hamilton et al. (2023) | United States | $n = 179$ (47% male, 53% female), NCAA DI (72%) and DIII (28%) student-athletes<br>Age: 21 ± 2<br>Tiers 3–4 | Last year | Supplement use: 97%<br>Supplements used by DI athletes: sports drinks, 86%; sports bars, 78%; protein shakes, 74%; recovery drinks, 70%; energy gel, 69%; multivitamins/minerals, 50%<br>Supplements used of DIII athletes: sports drinks, 79%; sports bars, 67%; protein shake, 54%; multivitamin/mineral, 53% | For DI and DIII athletes, respectively: preworkout, 19%/25%; BCAA, 9%/6%; herbs, 2%/7%; exotic berries, 2%/7%; Ashwagandha, 4%/4%; CBD, 2%/4%; Kava, 1%/1%; HMB 1%/0%; Ephedra, 1%/0%; Maca root, 1%/0%; Fenugreek, 1%/0%; Longjack, 1%/0% |
| Barrack et al. (2021) | United States | $n = 135$ (48% male, 52% female), NCAA DI collegiate runners<br>Age: 19 ± 0 years<br>Tier 4 | Last 4 weeks | Supplement use: 79%<br>Supplements used: multivitamin/mineral, 47%; iron, 47%; vitamin D, 34%; calcium, 33%; vitamin C, 16% | Amino acids, 3%; herbal supplements, 2% |
| Barrack et al. (2020) | United States | $n = 557$ (54% male, 46% female), NCAA DI collegiate athletes<br>Age: not defined<br>Tier 4 | Last year | Supplement use: 45%<br>Supplements used: protein powder, 21%; multivitamin/mineral, 13%; fish oil, 10%; creatine, 7%; vitamin C, 5% | Preworkout, 5%; herb/botanical supplements, 3% |
| Vento and Wardenaar (2020) | United States | $n = 138$ (36% male, 64% female), NCAA DI athletes<br>Age: 20 ± 2<br>Tier 4 | Last year | Supplement use: 100%<br>Supplements used: sports drinks, 94%; recovery drinks, 90%; energy bars, 89%; chocolate milk, 77%; protein shake, 66%; multivitamin/mineral, 65% | Weight gainer, 31%; BCAA, 22%; Maca root, 17%; Tribulus terrestris, 13%; HMB, 12% |
| Roy et al. (2021) | Canada | $n = 306$ (30% male, 70% female), varsity athletes from one university<br>Age: 21 ± 2 years<br>Tier 3 | Last 6 months | Supplement use: 58%<br>Supplements used: protein, 49%, vitamins and minerals, 48%; carbohydrate, 36% | For males and females, respectively: stimulants (not further defined), 31%/12%; amino acids, 22%/11%; weight gainers, 8%/6%; herbs/botanicals, 8%/1%; fat burners/weight loss, 3%/1% |

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**Table 1 *(continued)***

| Study | Country | Athlete population | Time frame | Nutritional supplement use (most frequent) | Supplements at risk[a] |
|---|---|---|---|---|---|
| Madden et al. (2018) | Canada | $n = 42$ (79% male, 21% female), provincial, national, and international level wheelchair rugby athletes<br>Age: 36 ± 10 years<br>Tiers 2–4 | Last 3 months | Supplement use: 91% (male) and 78% (female)<br>Supplements used: sport or electrolyte drinks, 56%; protein or sport bars, 44%; vitamin D, 41%; protein powder, 29%; multivitamin/mineral, 29% | Plant Extracts/herbal supplements, 14%; preworkout, 10%; BCAA, 10% |
| Wiens et al. (2014) | Canada | $n = 63$ (40% male, 60% female in total sample), athletes competing at a minimum of a club level (subsample from a study with $n = 567$ athletes, including athletes younger than 18 years)<br>Age: 19–25 years (range)<br>Tiers 3–4 | Current use | Supplement use: 100%<br>Supplements used: sport/electrolyte drinks, 92%; vitamin C, 81%; protein, 71%; multivitamin/mineral, 68% | Total study sample: plant/herbal extracts, 47%; BCAA, 8% |

*Note*. The time frame for each study is based on the period of the original investigation. BCAA = branch chain amino acids; CBD = cannabidiol; HMB = hydroxymethylbutyrate; NCAA = National Collegiate Athletic Association (United States); DI = Division I; DIII = Division III; IQR = interquartile range.
[a]"Supplements at risk" refers to products considered more likely to be contaminated or adulterated with substances prohibited by anti-doping agencies. These substances may also appear in the "most frequently consumed" column if applicable. Athlete Tiers: Tier 1: Recreationally active; Tier 2: Trained/ Developmental; Tier 3: Highly trained/national level; Tier 4 = Elite/international level; and Tier 5: World class (McKay et al., 2022).

stated to allow for anonymization of actual statements; only the region that provided each statement was mentioned. The left column states the anecdote; the right column gives a consensus statement from the authors.

# Results

## What Supplements Do Athletes Use per Region?

The use of nutritional supplements is a widespread practice among athletes (Table 1), motivated by the desire to enhance performance, improve recovery, and maintain overall health. A considerable body of research has investigated this phenomenon across all six global regions, encompassing more than 32 countries. Most of the studies presented focus on elite athletes, covering a wide spectrum of sports disciplines, including endurance, strength, team, and individual sports. Despite the breadth of research, there is significant variability in how self-reported supplement use is questioned and reported, complicating cross-study comparisons. This inconsistency arises from several methodological differences, including the time frame assessed (ranging from current use to lifetime use), the specific athlete populations surveyed (e.g., Olympic vs. national-level athletes vs. amateur athletes), and the design of data collection tools. For example, some studies use fixed predefined supplement checklists, while others employ open-ended questions. Additionally, not all studies provide a complete list of supplements used, and in some cases, products were grouped into broader categories, further limiting comparability. Additionally, not all regions have the same definition of what is and/or is not a supplement, so the results reflect some ambiguity, as studies may or may not have included sports foods, functional foods, and substances that can be defined as medical products.

Prevalence rates of supplement intake among athletes varied dramatically, ranging from as low as 7% to as high as 100%, with Western regions reporting at the higher end of the spectrum. Importantly, although the self-reported supplement use seems to be lower in non-Western regions, it is difficult to assign a specific percentage to each of these regions, as only a limited number of studies are available, and generalization needs to be done with caution, as results may not be representative for the complete area and population. Across studies and regions, the most reported supplements included protein powders, vitamins, minerals, carbohydrate-containing sports drinks, creatine, caffeine-based products, and amino acids. Supplement use and choice were influenced by multiple factors, including the type of sport, sex, age, nutritional knowledge, and cultural or regional practices.

## What Is the Use and Knowledge of Third-Party Tested Supplements Among Athletes?

Because third-party certification organizations are primarily located in Western countries (e.g., Europe, North America, and Australia, regions with shared political, economic, and cultural systems rooted in European traditions), a separate table was constructed to compile studies from all regions that reported the use of TPT or otherwise regulated nutritional supplements.

Table 2 provides an overview of studies that investigated the use and knowledge of TPT supplements among athletes. Most available studies investigating the use and knowledge of TPT nutritional supplements originate from the United States, where third-party certification programs such as NSF Certified for Sport and Informed-Sport are more established and widely recognized. Several professional sporting organizations have partnerships with these TPT programs and strongly encourage all teams and athletes to select nutritional supplement products that have been evaluated by a TPT organization. The four other published non-U.S. studies were conducted in Europe (Ireland, the Netherlands, and Spain) and Australia, while there were no studies identified from the other three regions. Despite a majority of athletes in Australia acknowledging supplements can cause positive doping violations (Dascombe et al., 2010), there has been limited exploration into the use of TPT supplements by athletes. A recent study (Sport Integrity Australia [SIA], 2025, in preparation) found that 41% of all athletes and 53% of National-/International-level athletes surveyed used only TPT supplements.

Reported use of TPT supplements varies across studies and regions, but consistently fails to approach all consumed supplements. Surveys of collegiate and professional athletes indicate that while many athletes use nutritional supplements regularly, the



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**Table 2 Third-Party Supplement Knowledge and Use of Athletes (Organized From Most Recent to Less Recent)**

| Study | Country | Athlete population | Time frame | TPT knowledge or use |
|---|---|---|---|---|
| Gimbar et al. (2025) | United States | $n = 118$ (47% male, 53% female), NCAA DI (67%), and DIII (33%) student-athletes<br>Age: 21 ± 2<br>Tiers 3–4 | Last year | 81% of athletes reported consuming some TPT nutritional supplement<br>TPT nutritional supplements: consistent (18%), inconsistent (57%), and no intake (25%)<br>TPT dietary supplements: consistent (48%), inconsistent (7%), and no intake (45%)<br>TPT sport foods: consistent (49%), inconsistent (3%), and no intake (57%) |
| Schott, Bhalla, et al. (2024) | United States | $n = 662$ (50% male, 50% female), NCAA DI student-athletes<br>Age: 20 ± 2<br>Tier 4 | Last year | 100% of the athletes reported the use of nutritional supplements<br>54% stated that they know where to find and order TPT supplements<br>77% recognized at least one TPT organization logo<br>TPT supplements: consistent (29%), inconsistent (71%) |
| Sport Integrity Australia (2025) (in publication; Study website) | Australia | $n = 879$ (59% male, 41% female) athletes from all level and sports. National/international level, $n = 164$ (16% of study)<br>Age: 15 and older<br>Tiers 3–4 | Last year | Total sample:<br>41% of athletes had used only TPT supplements<br>2% reported using some TPT supplements and some other supplements<br>20% reported not knowing whether or not they had used TPT supplements<br>19% reported being unfamiliar with TPT<br>For National/International athletes:<br>53% of athletes had used only TPT supplements<br>8% reported using some TPT supplements and some other supplements<br>15% reported not knowing whether or not they had used TPT supplements<br>10% reported being unfamiliar with TPT<br>29% reported being completely or fairly confident in explaining TPT to a teammate or parent/carer |
| Dow et al. (2025) | United States | $n = 86$ (60% male, 40% female), high school athletes<br>Age: 14–17 (range)<br>Tier 2 | Lifetime | 36% heard of the term "TPT"<br>16% stated that they knew where to find, and 15% knew where to order TPT supplements |
| Schott, Seltzer, et al. (2024) | United States | $n = 410$ (53% male, 47% female), NCAA DI student-athletes<br>Age: 21 ± 2 years<br>Tier 4 | Last year | 38% reported "consistent TPT use"<br>Supplements used: Electrolyte replacement beverages (72%), 67% TPT; Protein shakes (65%), 66% TPT; Recovery drinks (57%), 71% TPT; Caffeine (56%), 42% TPT; Creatine (22%), 79% TPT |
| Wardenaar, Lybbert, et al. 2024) | United States | $n = 225$ (72% male, 28% female), high school athletes<br>Age: 16 ± 1 years<br>Tier 2 | Last year | 94% of the athletes reported the use of at least one nutritional supplement<br>Supplements used: Vitamins (65%), 56% TPT; Protein powder (58%), 53% TPT; Creatine (23%), 67% TPT; Preworkout (20%), 53% TPT; All other supplements were reported to be third-party tested less often, ranging from 0% to 49% |
| McDaid et al. (2023) | Northern Ireland and Republic of Ireland | $n = 217$ (51% male, 49% female), international championship ($n = 55$) and amateur ($n = 162$) athletes<br>Age: 18 years and older<br>Tiers 2–4 | Current use | Between 20% (amateur) and 29% (elite) would refer to a "batch testing/TPT organization" such as "Informed Sport" before taking a botanical food supplement |

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**Table 2 *(continued)***

| Study | Country | Athlete population | Time frame | TPT knowledge or use |
|---|---|---|---|---|
| Wardenaar et al. (2020) | The Netherlands | $n = 601$ (50% male), Dutch Olympic status and non-Olympic status athletes<br>Age: 27 ± 10 years<br>Tiers 3–4 | Current use | 50% of the athletes stated that they were familiar with the Dutch TPT supplement system (NZVT). 75% of a subgroup ($n = 370$) reported that they use the NZVT list while purchasing products |
| USADA 2022<br>USADA (n.d.) | United States | $n = 994$ (44% male, 56% female), USADA Registered Testing Pool and Clean Athlete Pool<br>Age: undefined<br>Tiers 3–5 | Last year | 68% of the athletes reported the use of dietary supplements<br>56% reported always checking NSF certification before taking a supplement |
| Vento and Wardenaar (2020) | United States | $n = 138$ (36% male, 64% female) NCAA DI athletes<br>Age: 20 ± 2<br>Tier 4 | Last year | 66% purchased nutritional supplements not provided by the athletic department. Of those, 57% stated the supplements they bought were TPT |
| Baltazar-Martins et al. (2019) | Spain | $n = 527$ (66% male, 34% female), high-performance athletes<br>Age: <15 to >41 years<br>Tiers 3–4 | Last year | 85% of the sample (337 DS consumers) indicated that they only used certified supplements free of doping agents, 81% were not aware of any platform to verify supplement safety/quality<br>92% of the sample considered that supplements are safe and controlled by the supplement company/brand<br>40% of athletes had knowledge of a Spanish national-based application that certified permitted supplements and identified doping agents (i.e., NoDop App) |

*Note*. Time frame for each study is based on period of original investigation. Athlete tiers: Tier 1: Recreationally active; Tier 2: Trained/developmental; Tier 3: Highly trained/National level; Tier 4: Elite/International level; and Tier 5: World class (McKay et al., 2022). TPT = third-party testing; CBD = cannabidiol; HMB = hydroxymethylbutyrate; NCAA = National Collegiate Athletic Association (United States); DI = Division I; DIII = Division III; USADA = U.S. Anti-Doping Agency.

percentage of TPT supplements used ranges from 50% to 82% (Gimbar et al., 2025; McDaid et al., 2024; Schott, Seltzer, et al., 2024; Vento & Wardenaar, 2020; Wardenaar et al., 2020). When specifically checking TPT use for each supplement, only 18% to 34% of the athletes consistently report that all their supplements are TPT evaluated (Gimbar et al., 2025; Schott, Seltzer, et al., 2024; Wardenaar, Lybbert et al., 2024; Wardenaar, Schott et al., 2024).

Even among athletes who are aware of certification programs, actual adherence to using certified products was low. There was also confusion regarding the difference between supplements that are TPT and those simply marketed as "safe" or "natural," which may not provide any assurance against contamination or containing banned substances.

## What Legislation Exists for Each Region?

Most countries, as reported in Table 3, have a designated ministry tasked with health, often including food safety and consumption (sometimes part of agriculture); as such, each country can have its own definition of nutritional supplements. In addition, there is often an authority (Table 3) that regulates the manufacturing of nutritional supplements, as well as an entity that advises on questions about dietary recommendations and food safety. Nutritional supplements are often included in food law, and specifically, nutritional supplement legislation may overlap with legislation regarding medication. Only the European Union, covering the major part of Europe, has an overarching Legislation Directive 2002/46/EC for "food supplements."

Some countries are organized in larger regions, for example, Mercosur, in Latin America, promotes harmonization of laws and regulations across its member countries. In addition, some countries have specific regulations for sports nutritional supplements, such as Brazil (Aith et al., 2022). In Australia and New Zealand, sports supplements are specifically regulated in accordance with national legislation by government agencies. In New Zealand, nutritional supplements are regulated under the Dietary Supplements Regulations 1985 (Dietary Supplements Regulations 1985 (SR 1985/208) (as at 18 December 2024) Contents—New Zealand Legislation, n.d.), with these regulations currently under review. In Australia, nutritional supplements focusing on sports performance may be regulated as therapeutic goods (medicine) or food, depending on the claim. Changes to regulations in Australia were implemented between 2020 and 2023. In China, the government requires supplements to be sold with a blue cap as part of its national regulatory framework, which may increase clarity and visibility of nutritional supplements. However, to the author's knowledge, no other countries use this type of visual cue.

Out of 195 countries, more than 100 countries have incorporated the World Health Organization Good Manufacturing Practice (GMP) provisions into their national medicine laws. Next to this, many more countries have adopted its provisions and approach in defining their own national GMP requirements (Health Products Policy and Standards, n.d.). At the same time, countries can adhere to GMP or a Pharmaceutical Inspection Co-operation Scheme, without applying this directly to the production of nutritional supplements. A nonexhaustive list of examples with exceptions are the United States, mandating cGMP for dietary supplements (Code of Federal Regulations, n.d.); Australia, requiring GMP compliance for some supplements (complementary medicines); Canada, requiring all natural health products, including dietary supplements, be



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**Table 3 Legislation for Nutritional Supplements per Region**

| | Provide context | Resource |
|---|---|---|
| Africa | | |
| Harmonization | No, nutritional supplement regulation is not harmonized between African countries. | |
| Region-wide organization | No regional organization exists. | |
| Country-based examples | | |
| Egypt—Authorities | EDA: Oversees the regulation of pharmaceuticals, medical devices, and health supplements, ensuring that these products meet Egypt's rigorous health standards. | https://www.edaegypt.gov.eg |
| | NFSA: Manages the regulation of food products, ensuring they comply with the country's safety and quality standards. | https://nfsa.gov.eg/en-gb |
| | GOEIC: Responsible for the inspection and quality control of imported goods, including food products and industrial items, ensuring they meet Egyptian standards. | https://www.goeic.gov.eg/en |
| Egypt—Legislation | Law No. 1 of 2017: NFSA oversees national food safety, including dietary supplements. | https://nfsa.gov.eg/en-gb |
| South Africa—Authorities | SAHPRA: There is limited regulation (and enforcement) of products sold as "health supplements," considered to be Complementary Medicines. Regulation is limited to a paper-based evaluation of health claims made on these products (see definition); there is no actual testing of product formulations, nor the safety or efficacy of the products. | https://www.sahpra.org.za |
| South Africa—Legislation | Medicines & Related Substances Act 101 of 1965: Regulation of Medicines and the subcategory of Complementary Medicines (including "health supplements") in South Africa. | https://www.gov.za/documents/ |
| Morocco—Authorities | MHSP: Oversees the approval and monitoring of dietary supplements, ensuring they meet health and safety standards. | https://www.sante.gov.ma |
| | Collaborates with other ministries to establish regulations governing the marketing and distribution of these products.<br>National Office for Food Safety (ONSSA): Responsible for the safety and compliance of food products, including dietary supplements. Conducts health safety and conformity checks on plant-based products, food additives, and supplements at national levels, as well as for imports and exports. | https://www.onssa.gov.ma/e |
| Morocco—Legislation | Law No. 10-22; Decree No. 2.23.1055: Regulation of health products, including dietary supplements. | https://www.cg.gov.ma/en/10-22 |
| | Law No. 28-07; Decrees No. 2-10-473 & 2-12-389: Regulation of food safety. | https://www.onssa.gov.ma/28-07 |
| Asia | | |
| Harmonization | No, nutritional supplement regulation is not harmonized between Asian countries. | |
| Regionwide organization | Partly; GCC and the GSO are regional organizations which consists of the national standardization bodies of GCC member states. | https://www.gcc-sg.org/en<br>https://www.gso.org.sa/en/ |
| Country-based examples | | |
| Saudi Arabia—Authorities | SDFA: Regulate nutritional supplements (approving and registering supplements), functional foods, and health claims. They also regulate safety, efficacy, and labeling compliance, while monitoring imported and locally manufactured products. | https://www.sfda.gov.sa/en |
| Saudi Arabia—Legislation | Royal Decree No. (M/6) dated 25/1/1428H (February 13, 2007): officially established SDFA as an independent body, allowing it to regulate standards and procedures related to food safety. | https://www.fao.org/faolex/ |
| Japan—Authorities | MHLW: Regulates the safety, standards, and ingredient approvals of nutritional supplements to protect public health. | https://www.mhlw.go.jp/stf/e |
| | CAA: Oversees labeling and health claims on supplements, ensuring they are accurate and not misleading. | https://www.caa.go.jp/en/ |
| | FSC: Conducts scientific risk assessments of supplement ingredients to evaluate their safety for human consumption. | https://www.fsc.go.jp/e |
| | MAFF: Promotes the use and quality of agricultural ingredients in supplements and supports research into their functional properties. | https://www.maff.go.jp/e |

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**Table 3 *(continued)***

| | Provide context | Resource |
|---|---|---|
| Japan—Legislation | FSA: To ensure food and nutritional supplement safety. | https://www.mhlw.go.jp/ |
| | Food Labeling Act: To protect consumers from misleading claims, labeling requirements are specified. | https://www.caa.go.jp/en/ |
| | FOSHU: Health claims operated under the supervision of MHLW. | https://www.mhlw.go.jp/en |
| China—Authorities | NMPA: Regulates health food products, including dietary supplements. Previously known as CFDA.<br>SAMR: Overseeing dietary supplement market regulation, labeling, safety inspections, manufacturing, and advertising. | https://english.nmpa.gov.cn/ |
| China—Legislation | Drug Administration Law of the People's Republic of China (2019 Revision): Framework for the administration of drugs, including certain health foods with therapeutic claims. | https://english.nmpa.gov.cn/ |
| | Food Safety Law of the People's Republic of China (2015): General principles and requirements for food safety for health foods and dietary supplements. | https://www.chinalawtranslate.com/en/ |
| | Administrative Measures on Health Food Registration and Filing (CFDA Order No. 22): Procedures for registering health foods, including ingredients and claims. | https://www.cirs-group.com/en/ |
| Australia and New Zealand | | |
| Harmonization | No harmonization on dietary supplement regulation in the broader Oceania region. | |
| Regionwide organization | Yes, there is a partial regional organization via the FSANZ, which provides central scientific advice and support, while developing standards, including for Sports Foods and electrolyte drinks. FSANZ is also monitoring safety and supporting consumers. It does not regulate therapeutic products such as high-dose vitamins or herbal medicines. FSANZ also provides central scientific advice and support, while developing standards, monitoring safety, and supporting consumers. | https://www.foodstandards.gov.au/ |
| Country-based examples | | |
| Australia—Authorities | TGA: Regulates therapeutic goods such as high-dose vitamins, herbal supplements, and probiotics that come with health or medicinal claims. If a supplement makes claims for improvement or maintenance of physical or mental performance in sport or exercise or recreational activity, provided as a tablet, capsule, or pill, it is considered a therapeutic good. Additionally, products that contain substances from the WADA prohibited substance list or the Australian Poisons Standard are considered therapeutic goods. | https://www.tga.gov.au/ |
| | ACCC: Ensuring truthful advertising and labeling on nutritional supplements. | https://www.accc.gov.au/ |
| Australia—Legislation | Food Standards Code: If a Sports Supplement does not meet the definition of a therapeutic good, it is regulated as a food. Australian State or Territory agencies and the Department of Agriculture, Fisheries and Forestry enforce the Food Standards Code.<br>Therapeutic Goods Administration Act 1989: Framework for regulation of therapeutic goods, including certain nutritional supplements. | https://www.foodstandards.gov.au/food-standards-code<br>https://www.tga.gov.au/resources |
| New Zealand—Authorities | Ministry of Health: administers the Dietary Supplements regulations 1985 (revised version 2024). New Zealand Medicines and Medical Devices Safety Authority (Medsafe): is responsible for administering the Dietary Supplement regulations on behalf of the Ministry of Health.<br>Ministry for Primary Industries: administers the Food Act 2014, the Animal Products Act 1999, and the Biosecurity Act 1993. | https://www.legislation.govt.nz/1985<br>https://www.medsafe.govt.nz<br>https://www.mpi.govt.nz/ |
| New Zealand—Legislation | Dietary Supplement Regulations 1985 (revised version 2024): these regulations provide definitions of dietary supplements as well as specify requirements for some ingredients and include labeling and maximum permitted daily doses for several vitamins and minerals.<br>Dietary Supplements containing ingredients from animals and animal products must comply with the Food Act 2014, the Animal Products Act 1999, and the Biosecurity Act 1993.<br>Food Act 2014: administered by the Ministry for Primary Industries and ensures the provision of safe and suitable food (including supplements from animals and animal products) for sale in New Zealand.<br>Animal Product Act 1999: administered by the Ministry of Primary Industries and aims to minimize and manage risks to human and animal health arising from the production and processing of animal material and products.<br>Biosecurity Act 1993: administered by the Ministry for Primary Industries and aims to regulate the importation of goods (including Dietary Supplements) to protect New Zealand's environment, plants, and animals from unwanted pests and diseases. | https://www.legislation.govt.nz/1985<br>https://www.legislation.govt.nz/2014<br>https://www.legislation.govt.nz/1999<br>https://www.legislation.govt.nz/1993 |

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**Table 3 *(continued)***

| | Provide context | Resource |
|---|---|---|
| Europe | | |
| Harmonization | Yes, in Europe, there is harmonization under the EFSA, with its headquarters in Parma, Italy. The agency provides scientific advice on risks related to food and was established in 2002. | https://www.efsa.europa.eu/en |
| Regionwide organization | Yes, EFSA issues the independent Scientific opinion, which is then underpinned by EU law. Currently, EU law is retained in the United Kingdom, so they are harmonized. Sports foods are not singled out and classified as a food supplement or "regular food." Depending on how the product is manufactured, it must adhere to relevant legislation with respect to labeling and product composition (including product requirements for fortification, nutrition, and health claims). Athletes are not classified as a specific vulnerable group of consumers; and therefore, no extra legislation is available. The following key EFSA legislation focuses on nutritional supplements in the EU:<br>Directive 2002/46/EC—Food Supplements Directive: Regulates the composition and labeling of food and nutritional supplements.<br>Regulation (EC) No 1924/2006—Nutrition and Health Claims Regulation: To prevent misleading information, nutrition and health claims are regulated.<br>Regulation (EC) No 1925/2006—Regulation of the addition of vitamins, minerals, and other substances to foods, including food supplements.<br>Regulation (EC) No 178/2002—General Food Law: General principles of food law, including food safety.<br>Regulation (EU) 2015/2283—Novel Foods Regulation: Regulation of foods and new ingredients not commonly consumed within the EU before 1997.<br>Geographically, Russia is part of the European continent, but it does not align with EU legislation or EFSA. Russia operates its own legislation and engages in a separate regulatory framework, the EAEU, but none of the members are part of the EU. The RUSADA strongly recommends against the use of nutritional supplements because of the risk of inadvertent doping. | https://www.efsa.europa.eu/en<br>https://eur-lex.europa.eu/homepage<br>https://www.eaeunion.org/?lang=en<br>https://rusada.ru/en/substances/bad/ |
| Country-based examples | | |
| United Kingdom—Authorities | Department of Health and Social Care: Developing national nutrition policy for nutrition and food supplements in collaboration with other departments like the FSA, responsible for food safety, and MHRA, regulating supplements classified as medicines.<br>DEFRA: Overseeing broader food policy, including labeling.<br>Public Health England (now part of the UK Health Security Agency and Office for Health Improvement and Disparities): Provides guidance on nutrition and health, including supplementation. | https://www.gov.uk/government/DHSC<br>https://www.food.gov.uk<br>https://www.gov.uk/government/MHRA<br>https://www.gov.uk/government/DEFRA<br>https://www.gov.uk/government/HSA<br>https://www.gov.uk/government/HID |
| United Kingdom—Legislation | There is no specific legislation on foods intended to meet the expenditure of intense muscular effort, especially for sportspeople; therefore, general food law applies. Products presented as "food supplements" need to comply with the Food Supplements (England) Regulations 2003 and equivalent legislation in Scotland, Wales, and Northern Ireland. All products presented for sportspeople need to ensure that any nutrition or health claims made are compliant with retained Regulation (EC) No 1924/2006. | https://www.legislation.gov.uk/2003 |
| The Netherlands—Authorities | Ministry of Health, Welfare and Sport (VWS): Responsible for overall policy and legislation on public health, including setting broader frameworks for supplement regulation. | https://www.government.nl/VWS |
| | NVWA: The Dutch Food and Consumer Product Safety Authority monitors the safety and labeling of supplements, ensuring they meet food law requirements. | https://english.nvwa.nl/ |
| | CBG: The Medicines Evaluation Board steps in when supplements make medicinal claims or when products border between supplements and medicines. | https://english.cbg-meb.nl/ |
| | RIVM: The National Institute for Public Health and the Environment conducts research and advises the government on public health risks related to supplements. | https://www.rivm.nl/en |
| The Netherlands—Legislation | Warenwet (Commodities Act): The Warenwetbesluit Voedingssupplementen ("Commodities Act Decree on Food Supplements") specifically sets rules for vitamin and mineral supplements. | https://business.gov.nl/CA |
| | Medicines Act (Geneesmiddelenwet): Supplement marketed with a therapeutic (medical) claim may be classified and regulated as a medicine instead of a food or supplement. | https://wetten.overheid.nl/GMW |

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**Table 3** ***(continued)***

| | Provide context | Resource |
|---|---|---|
| Germany—Authorities | BMEL: The Federal Ministry of Food and Agriculture sets policy and oversees national food regulations, including for supplements. In Germany, nutritional supplements ("Nahrungsergänzungsmittel") are mainly regulated as foods, not medicines, and several authorities are involved. | https://www.bmel.de/EN/ |
| | BVL: The Federal Office of Consumer Protection and Food Safety monitors food supplements' safety and coordinates food monitoring programs. | https://www.bvl.bund.de/EN/ |
| | BfArM: The Federal Institute for Drugs and Medical Devices steps in if a supplement is considered a medicinal product based on its composition or claims. | https://www.bfarm.de/EN/ |
| | Local Food Control Authorities (Lebensmittelüberwachungsämter): Regional (state level) agencies inspect, control, and enforce food law compliance in retail, production, and online sales. | https://www.bmel.de/EN/ |
| | BfR: The Federal Institute for Risk Assessment evaluates health risks related to supplements and advises the government. | https://www.bfr.bund.de/en/ |
| Germany—Legislation | LFGB: Central German food law, also applying to nutritional supplements. | https://www.gesetze-im-internet.de/lfgb/ |
| | AMG: Medicines act when a supplement claims to have medicinal properties. | https://www.gesetze-im-internet.de/englisch_amg/ |
| | Verpackungsverordnung: This packaging ordinance lists requirements for labeling (including supplements). | https://www.lizenzero.de/en/packaging |
| Latin America | | |
| Harmonization | Yes, to some extent, based on the ALANUR, is an agency with representatives of 15 countries of the region. The goal of ALANUR is to join forces with the industry and authorities to provide reliable information regarding dietary supplements. ALANUR's mission is to assist in the creation of appropriate regulatory frameworks to guarantee responsible access to nutritional supplements in the Latin American market. ALANUR does not have legal force. | https://alanurla.org |
| Region-wide organization | Yes, there are three regional initiatives: | |
| | The Mercado Comum do Sul (Mercosur): (Argentina, Bolívia, Brazil, Paraguay, and Uruguay) is a South American regional intergovernmental organization founded by the Treaty of Asunción on March 26, 1991, which aims, among other things, to promote the free circulation of goods among its member countries. | https://www.mercosur.int/en/ |
| | Pacific Alliance: (Chile, Colombia, Mexico, and Peru), was established on April 28, 2011, and focuses on creating, in a participatory and consensual manner, an area of deep integration that progressively advances toward the free movement of goods, services, capital, and people. The alliance establishes requirements for the commercialization of food supplements, in terms of good manufacturing practices, certificates of free sale, sanitary registrations, labeling, and conditions of sale that must be complied with. | https://pacificallianceblog.com/ |
| | Tratado General de Integración Económica Centroamericana: (Guatemala, El Salvador, Nicaragua, Honduras, and Costa Rica), which was established in 1961 with the aim of reaffirming its purpose of unifying the economies of the four countries and jointly promoting the development of Central America. The treaty has the purpose of establishing the requirements that must be met by the nutritional labeling of pre-packaged food products for human consumption for the population from 3 years of age. | https://www.mineco.gob.gt/tratados-vigentes-int-economica-centroamericana |
| Country-based examples | | |
| Brazil—Authorities | ANVISA: Supplements are classified as foods, not medicines. ANVISA lists authorized ingredients and health claims, products must comply with regulations for labeling, safety, and composition standards. | https://www.gov.br/anvisa/pt-br |
| Brazil—Legislation | RDC Resolution No. 243/2018: Establishes the requirements for food supplements (i.e., products intended for oral ingestion, presented in pharmaceutical forms), focusing on composition and quality standards, labeling requirements, registration and notification, and compliance. | https://bvsms.saude.gov.br/No243 |
| | RDC No. 240/2018: Positive list of substances allowed in food supplements, including exempted substances (such as sugars, sweeteners, mineral water, certain diet-specific foods, and some supplements), mandatory registration for novel foods and ingredients, and products with health claims. | https://bvsms.saude.gov.br/No240 |
| | RDC No. 241/2018: Regulates inclusion of probiotics in supplements related to safety and health benefits. | https://bvsms.saude.gov.br/No241 |

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**Table 3 *(continued)***

| | Provide context | Resource |
|---|---|---|
| Colombia—Authorities | Ministerio de Salud y Protección Social: The Ministry of Health and Social Protection is responsible for public health policies, including the regulation of food, medicine, and dietary supplements. | https://www.minsalud.gov.co/ |
| | INVIMA: This is the regulatory body under the Ministry of Health and Social Protection, responsible for product registration and approval, monitoring, labeling and advertising control, import and export oversight, health and safety risk management, as well as consumer education. | https://www.invima.gov.co/ |
| Colombia—Legislation | Decree 3249 of 2006: Regulatory document for dietary and nutritional supplements. | https://www.minsalud.gov.co/3249 |
| | Resolution 333 of 2011: Describes the requirements for labeling and advertising. | https://scj.gov.co/R333 |
| | Decree 677 of 1995: This is a framework for health registration of medicines and related products, which may apply to certain supplements. | https://www.funcionpublica.gov.co/677 |
| Argentina—Authorities | Ministerio de Salud de la Nación (Ministry of Health): Sets national health policies, including nutrition-related health strategies and supervises ANMAT. | https://www.argentina.gob.ar/salud |
| | ANMAT: This regulatory agency is involved in nutritional supplement safety, including product registration and authorization, ingredient regulation, labeling and health claims, safety monitoring, recalls and alerts, regulating advertising, and consumer education. | https://www.argentina.gob.ar/anmat |
| Argentina—Legislation | CAA—Alimentos de regimen o dietéticos: Describes key aspects toward supplement safety, including composition, health claims, labeling requirements, safety, manufacturing standards, and product registration and control. | https://www.argentina.gob.ar/dieteticos |
| | Disposición ANMAT 4980/2005—Regulates "suplementos dietarios": Guidelines for advertising dietary supplements. It includes a statement that supplements may not claim therapeutic or medicinal properties. | https://www.anmat.gob.ar/4980 |
| | Disposición 1631/2011—Guidance on maximum nutrient limits: This disposition aims to provide clear and accurate information about dietary supplements, promoting safety and informed decision making. | https://www.rosario.gob.ar/1631 |
| Mexico—Authorities | Secretaría de Salud (Ministry of Health): Oversees national health policy and works in conjunction with COFEPRIS to regulate health products. | https://www.gob.mx/salud/en |
| | COFEPRIS: Regulatory body under the Ministry of Health responsible for the control and regulation of health products, including dietary supplements. | https://www.gob.mx/cofepris |
| Mexico—Legislation | General Health Law (Ley General de Salud): Legal framework for health regulation in Mexico, including provisions for dietary supplements. | http://www.diputados.gob.mx/ |
| | NOM-051-SCFI/SSA1-2010: Official Mexican standard for general labeling specifications for prepackaged food and nonalcoholic beverages, including dietary supplements. | https://www.dof.gob.mx/051 |
| | NOM-241-SSA1-2012: Establishes GMPs for establishments dedicated to the manufacture of medical devices, which can be applicable to certain dietary supplements. | https://www.dof.gob.mx/241 |
| North America | | |
| Harmonization | No, there is no unified regulatory framework in North America covering the United States of America, Canada (and Mexico). | |
| Region-wide organization | No, but the USMCA, which replaced NAFTA in 2020, the countries agreed to intensify regulation, science-based risk assessment, recognition of Codex Alimentarius (i.e., a joint body of the FAO and the WHO) food standards as an international benchmark, and reduce trade barriers for nutritional and functional foods. | https://ustr.gov/trade-agreements/ |
| Country-based examples | | |
| United States—Authorities | U.S. FDA: Regulating food safety, dietary supplements, drugs, and medical devices.<br>USDA: Overseeing food labeling and nutrition programs.<br>National Institute of Health—Office of Dietary Supplements: The NIH ODS provides a wide range of evidence-based resources and educational materials to help athletes and the general public make informed decisions about dietary supplements. Resources include the ODS website with information about a wide range of supplements, dietary supplement fact sheets, supplement label database, consumer warnings and safety alerts, and interactive tools and nutrient recommendations.<br>CDC: Tracks supplement-related adverse events in collaboration with the FDA. | https://www.fda.gov<br>https://www.usda.gov<br>https://ods.od.nih.gov<br>https://ods.od.nih.gov/Supplements<br>https://www.cdc.gov |

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**Table 3** ***(continued)***

| | Provide context | Resource |
|---|---|---|
| United States—Legislation | DSHEA of 1994: This law defines dietary supplements and regulates their labeling, ingredients, and manufacturing. Supplements are categorized separately from drugs; and therefore, they do not require FDA preapproval before marketing, but manufacturers are responsible for ensuring supplement safety and truthful labeling. If an ingredient was not marketed in the United States before October 15, 1994, it must either be approved by the FDA through the NDI or GRAS notification process, or be determined as GRAS by an independent panel of qualified experts ("self-GRAS") before it can be legally sold. | https://ods.od.nih.gov/About/DSHEA |
| | FD&C Act: Authorizes the FDA to regulate nutritional supplements under DSHEA. The FDA may enforce action against unsafe or misbranded supplements. | https://www.fda.gov/FDC |
| | NLEA of 1990: Requires nutrition labeling on most packaged foods and nutritional supplements, which regulates nutrient content claims (e.g., "low fat," "high in vitamin C"). | https://www.congress.gov/NLEA |
| | FSMA of 2011: Strengthens FDA authority to prevent food contamination and includes supplement manufacturers under certain provisions. | https://www.fda.gov/FSMA |
| Canada—Authorities | Health Canada: Oversees the regulation of NHPs, including dietary supplements, ensuring they are safe, effective, and of high quality. | https://www.canada.ca/en/HC |
| | NNHPD: A division of Health Canada responsible for the implementation of the Natural Health Products Regulations, including the issuance of product licenses and site licenses. | https://www.canada.ca/NNHPD |
| | CFIA: Enforces food safety and labeling regulations, including those applicable to dietary supplements classified as foods. | https://inspection.canada.ca/en |
| Canada—Legislation | Natural Health Products Regulations (SOR/2003-196): Governs the sale of natural health products, including requirements for product licensing, labeling, and GMPs. | https://laws-lois.justice.gc.ca/2003 |
| | Food and Drugs Act: The primary legislation for the regulation of food and drugs, including provisions applicable to dietary supplements. | https://laws-lois.justice.gc.ca/FDA |
| | Food and Drug Regulations (C.R.C., c. 870): Provides detailed requirements under the Food and Drugs Act, including standards for foods for special dietary use. | https://laws-lois.justice.gc.ca/FDR |

*Note*. Date of access for all websites in this table is October 14th, 2025. ChatGPT (OpenAI) was employed to assist in the preliminary step of populating information for the context fields, this content was subsequently reviewed and verified. After these steps the authors, wrote up the table content. WADA = World Anti-Doping Agency; GMP = Good Manufacturing Practice; EDA = Egyptian Drug Authority; NFSA = National Food Safety Authority; GOEIC = General Organization for Export and Import Control; SAHPRA = South African Health Products Regulatory Authority; MHSP = Ministry of Health and Social Protection; GCC = Gulf Cooperation Council; GSO = GCC Standardization Organization; SDFA = Saudi Food and Drug Authority; MHLW = Ministry of Health, Labour, and Welfare; CAA = Consumer Affairs Agency; FSC = Food Safety Commission; MAFF = Ministry of Agriculture, Forestry and Fisheries; FSA = Food Sanitation Act; FOSHU = Food for Specified Health Uses; NMPA = National Medical Products Administration; CFDA = China Food and Drug Administration; SAMR = State Administration for Market regulation; FSANZ = Food Standards Australia New Zealand; TGA = Therapeutic Goods Administration;: ACCC = Australian Competition and Consumer Commission; EFSA = European Food Safety Authority; EU = European Union; EAEU = Eurasian Economic Union; RUSADA = Russian Anti-Doping Agency; FSA = Food Standards Agency; MHRA = Medicines and Healthcare products Regulatory Agency; DEFRA = Department for Environment, Food & Rural Affairs; NVWA = Nederlandse Voedsel- en Warenautoriteit; CBG = College ter Beoordeling van Geneesmiddelen; RIVM = Rijksinstituut voor Volksgezondheid en Milieu; BMEL = Bundesministerium für Ernährung und Landwirtschaft; BVL = Bundesamt für Verbraucherschutz und Lebensmittelsicherheit; BfArM = Bundesinstitut für Arzneimittel und Medizinprodukte; BfR = Bundesinstitut für Risikobewertung; LFGB = Lebensmittel- und Futtermittelgesetzbuch; AMG = Arzneimittelgesetz; ALANUR = Latin American Alliance for Responsible Nutrition; ANVISA = Agência Nacional de Vigilância Sanitária; INVIMA = Instituto Nacional de Vigilancia de Medicamentos y Alimentos; ANMAT = Administración Nacional de Medicamentos, Alimentos y Tecnología Médica; CAA = Código Alimentario Argentino; COFEPRIS = Comisión Federal para la Protección contra Riesgos Sanitarios; USMCA = United States-Mexico-Canada Agreement; FAO = Food and Agriculture Organization; WHO = World Health Organization; FDA = Food and Drug Administration; USDA = United States Department of Agriculture; NIH = National Institutes of Health; ODS = Office of Dietary Supplements; CDC = Centers for Disease Control and Prevention; DSHEA = Dietary Supplement Health and Education Act; NDI = New Dietary Ingredient; GRAS = Generally Recognized as Safe; FD&C Act = Food, Drug, and Cosmetic Act; NLEA = Nutrition Labeling and Education Act; FSMA = Food Safety Modernization Act; NHPs = natural health products; NNHPD = Natural and Non-prescription Health Products Directorate; NNHPD = Natural and Non-prescription Health Products Directorate; CFIA = Canadian Food Inspection Agency; NAFTA = North American Free Trade Agreement.

manufactured in compliance with GMP (Good Manufacturing Practices Guide for Drug Products (GUI-0001)—Canada.Ca, n.d.); Brazil requires GMP compliance for all companies producing nutritional supplements (Good Manufacturing Practices — Agência Nacional de Vigilância Sanitária—ANVISA, n.d.); South Africa requires compliance with GMP for products sold as health supplements (South African Health Products Regulatory Authority, 2025); China requires licensed manufactures to apply GMP (National Medical Products Administration, 2019); and GMP is required for manufacturers in Taiwan (Food and Drug Administration, Department of Health, Executive Yuan, 2013). In addition, there are countries that advocate the use of GMP, such as Japan, but currently only half of the manufacturers have adopted this policy (Sato et al., 2020).

## Which Systems Are Available to Ensure Low-Risk Supplement Choices?

### *Market-Driven: TPT Organizations by Region*

TPT organizations certifying products that have been tested for banned substances are operating out of Australia, Europe, and North America. Although some other regions hold local laboratories that



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can test supplements for contaminants and banned substances, they are not widely recognized or maintain a sufficient level of regulatory oversight. Access to international TPT organizations in Africa, Asia, and Latin America is often limited. Table 4 provides an overview of well-established TPT organizations that test supplements for banned substances. Table 5 lists additional well-established TPT organizations that assess product quality but do not screen for banned substances. As such, these are less suitable for athletes seeking to reduce anti-doping risk. It is important to mention that there are more organizations as listed in Tables 4 and 5; however, only well-established organizations are listed. Next, available TPT organizations per region are described in the following text below.

**Africa.** No national TPT organizations suitable for high-level athletes exist in Africa. Active marketing exists for TPT organizations such as Informed Choice and Informed Sport; however, only a limited number of local manufacturers make use of TPT services in general, and specifically, batch-testing for banned substances, due to high cost. Therefore, TPT products are usually only available from the European and American markets. In the end, the cost of these products becomes prohibitive for most local athletes.

**Asia.** In Asia, no widely recognized TPT organizations that test nutritional supplements for banned substances exist. Therefore, most sport organizations and athletes rely on international TPT organizations. The initiative for testing and education comes from national Olympic committees and NADOs. For example, the China Anti-Doping Agency website lists the option for paid testing of both foods and nutritional supplements, including, for example, for meat contamination. Batch testing costs roughly equate to $1,800 USD at present (Year 2025).

**Australia and New Zealand.** In Australia and New Zealand, the two main programs providing third-party batch testing are Human and Supplement Testing Australia (HASTA) and Informed Sport. HASTA is based in Australia, whereas Informed Sport is United Kingdom and/or United States-based. Both programs test each batch of manufactured products that participate in the program, including testing for banned substances. Some sport organizations, teams, or individuals may choose to have specific batches of sports supplements tested for their own use when a certain product is not readily available.

**Europe.** In Europe, a wide range of TPT organizations exist. Informed Sport and Informed Choice are based in the United Kingdom. In the Netherlands, the NZVT system (Dutch Safeguards System for Dietary Supplements in Elite Sports) is available with slightly under 500 tested products, although it mainly targets Dutch-speaking athletes; however, its website can be accessed in English. The German Cologne List (Kölner Liste) is accessible in German, English, and Spanish.

**Latin America.** There are no widely recognized TPT organizations based in Latin America that test supplements for banned substances. While some TPT-certified brands have entered the market through national distributors, international companies have at times faced legislative barriers—for example, in Brazil, where regulations have limited their ability to sell supplements. Local supplement brands may highlight quality control, GMP, and regulatory compliance (e.g., with Agência Nacional de Vigilância Sanitária), but they rarely emphasize TPT. A few isolated manufacturers offer certified products, though these certifications typically focus on product purity–such as oil quality—rather than screening for substances prohibited in sport.

**North America.** Many of the North American organizations operate globally and provide services to athletes and manufacturers in multiple regions. A wide range of TPT organizations that test supplements for banned substances are available to athletes in North America, including NSF Certified for Sport, Informed Choice, Informed Sport, and Banned Substances Control Group. Additional certification organizations—such as NSF, USP, Consumer Labs, and Labdoor—focus on product quality but do not test for banned substances in most cases. Notably, many of these North American organizations operate globally and provide services to athletes and manufacturers in multiple regions. However, certain organizations (e.g., Labdoor) offer selected certifications (e.g., "Sport") that include screening for banned substances, whereas others (e.g., "Quality") do not. Therefore, athletes and practitioners should carefully consider both the certifying organization and the specific certification type when selecting dietary supplement products, making sure that the selected product is TPT for banned substances. It should also be noted that across regions, certification programs differ markedly in their postcertification surveillance. For instance, whether they perform ongoing ("blind") retail testing and whether certification applies to every production lot or only selected lots. Accordingly, athletes and coaches should verify that the specific batch they plan to use has been tested for banned substances, particularly for programs that do not test every batch (e.g., Informed Choice). In some instances, this can be done by scanning the product barcodes with the organization's app (e.g., NSF certified for Sport). However, others may require more work by needing to locate the specific lot number on the container and cross-reference it with the listed tested lots on the organization's website (e.g., Informed Choice). Although the risk is lower for TPT organizations that test every production batch (e.g., Informed Sport), or allow the placing of their logo when manufacturers test all batches (e.g., NZVT), it is still recommended that athletes confirm that the exact lot in hand has been screened. Finally, only Informed Sport lists storing sealed containers of tested lots for the listed shelf life of the product after testing. Therefore, to support product traceability, athletes are advised to retain supplement packaging or record the lot number of any product used. Maintaining this information may aid verification in the event of a product recall, adverse event, or potential anti-doping inquiry (Eichner et al., 2019).

### *Supplement Education*

In most countries, NADOs and NOCs are responsible for educating athletes and support personnel about low-risk supplement use. Globally, WADA also provides a comprehensive education program. In many Western countries, national sports or performance nutrition associations help uphold professional standards and support informed low-risk decision-making around supplement use. Larger international and national organizations also play a key role in educating the broader public on performance nutrition. These include the International Olympic Committee and professional associations such as Professionals in Nutrition for Exercise and Sports, American College of Sports Medicine, National Athletic Trainers' Association, the National Strength and Conditioning Association, and the International Society for Sports Nutrition (ISSN), which provide evidence-based guidance for practitioners working with athletes. Underneath, the available educational organizations and efforts per region are described.

**Africa.** Africa currently has 18 recognized NADOs, and two RADO zones for countries without a NADO (WADA, n.d.). For example, the South African Institute for Drug-Free Sport (SAIDS) plays a key role in anti-doping education, including low-risk supplement use, through programs that target athletes and support



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**Table 4 Well-Established TPT Organizations Testing for Banned Substances**

| Organization | Headquarters | Number of products listed | Number of screened WADA prohibited substances | Are all batches tested? | Product sourcing | Toxicology screenings | How to access product testing results | Label verification | Website |
|---|---|---|---|---|---|---|---|---|---|
| NSF certified for sport | United States | 290+ | 290+ | No[b] | M | Yes | W, A | Yes | https://www.nsfsport.com/ |
| Informed choice[a] | United Kingdom | 200+ | 280+ | No (monthly) | R | Yes | W | No | https://choice.wetestyoutrust.com/ |
| Informed sport[a] | United Kingdom | 3000+ | 280+ | Yes | M[c] | Yes | W, A | No | https://sport.wetestyoutrust.com/ |
| Banned substances control group | United States | 450+ | 400+ | Yes | M[c] | Yes | W | Yes (only BSCG quality) | https://www.bscg.org/ |
| Labdoor Sport | United States | 4 | 280+ | No | R | Yes | W, A | Yes | https://labdoor.com/ |
| HASTA | Australia | 100+ | 250+ | Yes | M | No | W, A (Through the Sports Integrity App) | No | https://hasta.org.au/ |
| NZVT | Netherlands | 160+ | 160+ | No[b] | M | No | W | No | https://www.dopingautoriteit.nl/nzvt |
| Kölner Liste | Germany | 1,000+ | 200+ | No | M | No | W, A | Yes | https://www.koelnerliste.com/en/ |

*Note*. WADA = World Anti-Doping Agency; GMP = good manufacturing practices; HASTA = Human and Supplement Testing Australia; NZVT = Dutch Safeguards System for Dietary Supplements in Elite Sport; M = the product for the initial certification is provided directly from the manufacturer of the product; R = the product for the initial certification is sourced by purchasing it from online and in person retail stores; A = Organization supplied app; W = website.

[a]These brands are part of LGC Limited. [b]Manufacturers have the option to test all lots or only selected lots when applying for certification. Some organizations allow tested batches to carry their logo (e.g., NSF Certified for Sport), whereas others only allow adding the logo to the label when all batches are tested (e.g., NZVT). [c]Random off-the-shelf retail sampling post certification may occur at the discretion of the testing organization. The date of access for all websites in this table is October 14th, 2025. ChatGPT (OpenAI) was employed to assist in the preliminary step of populating information for the context fields, this content was subsequently reviewed and verified. After these steps, the authors wrote up the table content. All organizations use ISO 17025-accredited methods.



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**Table 5 Well-Established Organizations Certifying Product Quality Without Testing for Banned Substances**

| Organization | Headquarters | Number of products listed | Number of screened WADA prohibited substances | Require GMP certification | Website |
|---|---|---|---|---|---|
| USP Verification | United States | Not publicly listed | No | Yes | https://www.usp.org/verification-services/ |
| NSF | United States | Not publicly listed | No (unless Certified for Sport) | Yes | https://www.nsf.org |
| Consumer Labs | United States | 1,000+ | No | No | https://www.consumerlab.com/ |
| Bureau Veritas | France | Not publicly listed | No | Yes | https://www.bureauveritas.com/ |
| Eurofins | Luxembourg | Not publicly listed | No | Yes | https://www.eurofins.com/ |
| Intertek | United Kingdom | Not publicly listed | No | Yes | https://www.intertek.com/ |
| Labdoor | United States | 100+ | No (Unless Sport) | No | https://www.labdoor.com/ |
| QAI | United States | Not publicly listed | No | Yes | https://www.qai-inc.com/ |

*Note*. ChatGPT (OpenAI) was employed to assist in the preliminary step of populating information for the context fields, this content was subsequently reviewed and verified. After these steps, the authors wrote up the table content. All organizations apply toxicology screenings, label verification, and use ISO 17025-accredited methods. Date of access for all websites in this table is October 14th, 2025. QAI = Quality Assurance International; WADA = World Anti-Doping Agency; GMP = good manufacturing practices; USP = United States Pharmacopeia.

personnel at all levels of sport. Another example is the Egyptian Anti-Doping Organization, which conducts educational outreach for national sports federations and clubs on safe supplement use and doping risks. In Morocco, the Moroccan Anti-Doping Agency collaborates with the Olympic committee and sports federations to deliver nationwide campaigns for athletes, coaches, and support personnel. These efforts include workshops and seminars at sports clubs, universities, and training centers, informational materials in Arabic and French; education on the dangers of doping and the WADA Prohibited List; and guidance on choosing safe, TPT supplements.

**Asia.** In Asia, there are 32 recognized NADOs and three RADO zones (WADA, n.d.). Some examples of robust education programs exist in Japan (Japan Anti-Doping Agency) and South Korea (Korea Anti-Doping Agency). The Japan Anti-Doping Agency identified that 30% of anti-doping rule violations are linked to nutritional supplement use (Tabata et al., 2020). As a result, the Japan Anti-Doping Agency has invested in educating athletes on the risk of contaminated supplements and the importance of verifying supplement risk level. In addition, the Olympic Council of Asia has implemented a comprehensive regional education program using WADA's Anti-Doping Education and Learning platform, which focuses on raising awareness of supplement contamination and promoting informed decision-making (Anti-Doping Education and Learning, n.d.). Another example, to reflect outreach numbers, has been provided by the China Anti-Doping Agency, which conducts various types of education programs illustrated in their annual reports (CHINADA, 2025). For example, in 2023, 800 qualification programs and 800 outreach programs were done, along with ~1,700 education lectures on provincial- or national-level athletes.

**Australia and New Zealand.** Both countries have their own NADO. In Australia, education for athletes, support personnel, medical personnel, and parents is provided online and in person by SIA. Education about supplements is also provided through outreach sessions at sporting events and broader communications. SIA, the National Institute Network, including the Australian Institute of Sport, Sport Dietitians Australia, and National Sporting organizations, collaborate to ensure alignment of information and advice and reach across the sport system. SIA has developed a mobile app that enables athletes and support personnel to search for batch-tested supplements, verify whether specific products have been tested, receive warnings about known prohibited ingredients, and assess the risk level of untested products (Sport Integrity Australia, 2025). The combination of ongoing supplement education for athletes and medical support personnel (including physicians, dietitians, and pharmacists), storytelling by athletes with lived experience, the availability of the mobile app, and strong regulatory oversight has led to a significant decline in supplement-related doping violations. Between July 2016 and June 2017, 17 positive doping tests were linked to supplement use among approximately 4,000 athletes. In recent years, despite similar testing volumes, this number has dropped to almost zero (Australian Government, 2022).

Launched in 2014, the Australian Institute of Sport's Sports Supplement Framework was designed to help athletes and practitioners make informed decisions about supplement use. Central to this framework is the ABCD Classification System. This system ranks sports foods and supplement ingredients into four categories based on scientific evidence and practical considerations, including safety, legality, and performance benefits. In which A = strong scientific evidence for use in specific situations in sport using evidence-based protocols, B = emerging and/or mixed scientific support, deserving of consideration in specific populations or situations, C = scientific evidence not supportive of benefit among athletes or no research undertaken to guide an informed opinion, D = banned or at high risk of contamination with substances that could lead to an ADRV classification system. This system has since become a benchmark supplementation practices across a wide range of sports, including sailing (Caraballo et al., 2020), cycling (García-Durán et al., 2024; Valiño-Marques et al., 2024), football (Günalan et al., 2022; Sousa-Rufino et al., 2025), swimming (Jiménez-Alfageme, Domínguez, Sanchez-Oliver, et al., 2022; Moreno et al., 2022), running (Jiménez-Alfageme, Domínguez, Sánchez-Oliver, et al., 2022; Jiménez-Alfageme, Rubio-Quintanilla, et al., 2023), triathlon (Jiménez-Alfageme, Martínez-Sanz, et al., 2023), fencing (Mata et al., 2021), and handball athletes (Muñoz et al., 2020; Romero-García et al., 2024), and officiating (Sánchez-Oliver et al., 2024).



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In New Zealand, the Sport Integrity Commission Te Kahu Raunui, National Sporting Organizations, and High-Performance Sport New Zealand collaborate to deliver consistent, up-to-date education on supplement use and safety. This education is delivered both in-person and online. One of the key tools is an online Supplement Decision-Making Guide, which walks users through various considerations for safe supplement use. Athletes supported by High-Performance Sport New Zealand also have access to a Supplements Program. Participation requires a consultation with a performance nutritionist, who provides individualized recommendations for TPT supplements—often accompanied by a discount code. Athletes are responsible for purchasing and tracking their own supplements and can only join the program after completing supplement training and signing a disclaimer.

In recent years, New Zealand has reported a low number of supplement-related doping cases (DFSNZ, 2025). This low incidence may reflect the impact of structured education, mandatory supplement disclaimers that reinforce personal responsibility, improved access to TPT products, and strict adherence to the High-Performance Sport New Zealand Code of Conduct. The code specifies that only medical doctors and performance nutritionists are permitted to offer supplement advice to athletes, coaches, or support staff.

**Europe.** Nearly all European countries have a well-functioning NADO; therefore, there are 47 NADOs and no RADOs (WADA, n.d.). These NADOs provide education on supplement risks to athletes and support personnel. Not all European countries have established performance nutrition networks, but one example is the United Kingdom-based Sports and Exercise Nutrition Register. This voluntary register accredits qualified professionals who are deemed competent to work independently with performance-oriented athletes, as well as those engaged in sport, exercise, or physical activity for health. Several other European countries also have professional groups dedicated to sports nutrition within the dietetic profession. These include the Netherlands (Dutch Sports Dietitians Association), Finland (Finnish Sports Nutrition Network), Sweden (Swedish Association of Sports Nutrition, part of national dietetic organization), Germany (German Institute of Sports Nutrition and affiliated professional groups), Norway (Norwegian Dietetic Association, with specialization in sports nutrition), Austria (Sports Nutrition Working Group, part of the national nutrition society ÖGE), and Switzerland (Swiss Association of Sports Nutrition, in cooperation with national sports bodies). These organizations and their members collaborate with NADOs, NOCs, and athletes to promote low-risk supplement use.

**Latin America.** There are currently 38 NADOs and two RADOs (PAN-RADO) in Latin America and the Caribbean (WADA, n.d.). Although many Latin American countries align with WADA guidelines, Brazil and Colombia have taken additional legislative steps to address unintentional doping. In Colombia, products containing substances on the WADA Prohibited List must carry a warning label stating: "This product contains substances prohibited in sport" (DECRETO 3249 DE 2006, n.d.). In Brazil, more recent regulations prohibit the use of any WADA-prohibited substances in the composition of food supplements (Ministério Da Saúde Agência Nacional de Vigilância Sanitária RESOLUÇÃO DA DIRETORIA COLEGIADA-RDC No 243, DE 26 DE JULHO DE 2018, n.d.). There is no region-wide network in Latin America dedicated exclusively to sports nutrition education. However, several countries maintain their own national organizations. Country-specific associations include the Brazilian Sports Nutrition Association (Associação Brasileira de Nutrição Esportiva) and the Mexican Sports Nutrition Federation (Federación Mexicana de Nutrición Deportiva), which offer education in sports nutrition.

**North America.** Both the United States of America (USADA) and Canada (Canadian Centre for Ethics in Sport, CCES) have their own NADO (WADA, n.d.). The U.S. Anti-Doping Agency (USADA) provides a wealth of resources for helping athletes understand the safety, risks, and regulations surrounding nutritional supplements, such as the USADA Supplement Guide (Supplement Guide Reducing Supplement Risk, n.d.). Another example is USADA's Supplement 411 campaign, which educates athletes about the risks of nutritional supplements—including how they are regulated, how contamination can occur, and which products pose the greatest risk. The campaign also provides guidance on evaluating supplement labels, ingredient lists, and TPT certifications (Home | U.S. Anti-Doping Agency [USADA], n.d.). The U.S. Olympic Committee provides educational resources and guidance to help athletes make informed decisions about nutritional supplements while ensuring compliance with anti-doping regulations (USOPC | Nutrition, n.d.).

The National Institutes of Health Office of Dietary Supplements provides a wide range of evidence-based resources and educational materials to help athletes and the public make informed decisions about dietary supplements (About ODS, n.d.). Key resources include the National Institutes of Health Office of Dietary Supplements website, which provides both scientific and consumer-friendly information on nutritional supplements. Notable tools include a wide range of Dietary Supplement Fact Sheets (National Institutes of Health, n.d.-a) and the Dietary Supplement Label Database (DSLD, n.d.), a searchable resource that allows users to explore ingredient lists, dosages, and product labels—helping to identify potentially unsafe or misleading products (National Institutes of Health, n.d.-b). In addition to educational tools, the Office of Dietary Supplements issues consumer warnings and safety alerts, and collaborates with researchers to study the effects of nutritional supplements on athletic performance and recovery—offering evidence-based insights on products such as protein powders, creatine, caffeine, and amino acids. The U.S. Food and Drug Administration also provides educational resources on supplement regulation, labeling, safety, and adverse event reporting, and regularly issues alerts about products containing hidden or illegal ingredients.

Finally, there are other commercial organizations, aside from TPT organizations, that provide meaningful resources. These organizations include the Drug Free Sport International and the Gatorade Sports Science Institute, which provide a variety of resources and educational materials for athletes interested in nutritional supplements. There is also examine.com, a well-respected, independent resource that provides evidence-based information on the safety and efficacy of nutritional supplements for athletes.

In Canada, there are several organizations that focus on ensuring the safe and low-risk use of nutritional supplements and providing educational information for athletes and the public. Health Canada is the federal agency responsible for national health policy and the regulation of nutritional supplements (Government of Canada, n.d.-b). Under the Natural Health Products Regulations, all natural health products—including vitamins, minerals, herbal remedies, and amino acids—must obtain a Natural Product Number before they can be sold (Government of Canada, n.d.-a). This system is intended to ensure product safety, efficacy, and quality. Additionally, Health Canada introduced the



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Supplemented Foods Regulations. These regulations provide a framework for prepackaged foods containing added ingredients such as caffeine, vitamins, and minerals—aimed at helping consumers make informed choices (Government of Canada, n.d.-c). The CCES is Canada's national anti-doping authority, overseeing the Canadian Anti-Doping Program (Canadian Centre for Ethics in Sport, n.d.), which underscores the risks of supplement use, and highlighting that athletes remain fully responsible for any prohibited substance in their system. They recommend that athletes should only use TPT products. CCES also delivers educational content covering banned substances, testing procedures, and athletes' rights and obligations under Canadian anti-doping rules (Online Learning | Canadian Centre for Ethics in Sport, n.d.). There is also the Canadian Olympic and Paralympic Sport Institute Network (Canadian Sport Institute Pacific, n.d.) and National Sport Organizations, which consist of regional sport institutes or centers and national sport organizations across Canada that employ or collaborate with high-performance sport dietitians. The Canadian Olympic and Paralympic Sport Institute Network has recently published an internal Operational Best Practices document to guide all integrated sport science team members and athletes on safe/low-risk and best practice use of sport supplements. This document details everything from sponsorship deals with athletes and supplement companies to guidance on monitoring and tracking their supplement use. When athletes are onboarded, they must agree to abide by the recommendations put forth. Additionally, sport dietitians educate athletes on continued safe supplement use, which aligns with CCES guidelines and promotes recommended third-party-certified products to minimize doping risks. Collectively, these organizations and resources contribute to the promotion of safe nutritional supplement use in Canada through regulation, education, and awareness of available TPT programs and products in Canada.

### *Limitations to Access and Solutions for Low-Risk Access of Nutritional Supplements*

Research suggests that influencing athlete behavior requires more than education alone (Backhouse, 2023). For example, teams and/or organizations providing certain supplements to athletes, reducing the chance that they will purchase uncertified products themselves, can be effective in reducing the risk of an anti-doping rule violation (Shaw et al., 2016a). A key limitation is the lack of direct access to TPT organizations in many regions. The Global Drug Reference Online provides athletes and support personnel with information about the status of specific medications and ingredients based on the current WADA Prohibited List (Global Drug Reference Online, n.d.). However, it only covers medications from a limited number of countries (i.e., the United States, Canada, the United Kingdom, Switzerland, Australia, New Zealand, and Japan) and does not include supplements. Still, it can be a useful tool for checking whether specific ingredients found in supplement formulas appear on the WADA prohibited list. Other countries have similar tools for checking medications. The limitations to access and potential solutions by region are presented below, though the most significant barriers are often economic, regulatory, and logistical in nature.

Possible solutions include adopting a food-first strategy while working to improve access to safer, tested supplements. This could involve building a trusted vendor network, establishing shipping partnerships (potentially supported by local or regional subsidies), and creating regional institutes to facilitate certification across multiple countries. Next, potential limitations to access nutritional supplements and solutions for low-risk access to supplements per region are described below.

**Africa.** The limited availability of batch-tested products and the cost related to such products are prohibitive for athletes. A lack of knowledge about where to purchase these products also presents a barrier—from limited awareness of what is available to confusion about what TPT certification entails. This includes differences between programs (e.g., Informed Choice vs. Informed Sport) and misleading marketing practices, such as promoting brand-level certification when only a single product has been tested. The South African Institute for Drug-Free Sport includes guidance on lower-risk strategies in its education program, such as the use of TPT products and how to interpret TPT certifications.

In general, African athletes often face difficulty accessing low-risk TPT nutritional supplements. For example, the Moroccan market for nutritional supplements is not strictly regulated, allowing many unverified or counterfeit products to be sold freely, especially online and in gyms and informal shops. Internationally certified supplements are expensive and hard to find in the local market. Many athletes, especially at the amateur or semipro level, cannot afford or access these low-risk options, pushing them toward cheaper, potentially unsafe alternatives. In addition, many athletes do not have access to sports nutritionists or qualified healthcare providers who can advise on low-risk supplement use. Instead, they rely on advice from gym trainers, peers, online sources, or social media, which may increase the risk of misuse or reliance on unsafe products. Morocco is still developing a nutrivigilance system (a national system to monitor and regulate the safety of nutritional supplements). This creates a gap in market oversight, with few mechanisms to report or recall harmful products.

**Asia.** One solution applied by the China Anti-Doping Agency is conducting random testing of nutritional supplements. Test results are issued in annual reports. However, because formulas and manufacturing processes can change, athletes currently have no way to confirm whether other batches are also free of banned substances. At the national level, sports organizations purchase and distribute supplements to athletes. These organizations include research and clinical staff, such as dietitians, who are required to complete anti-doping training.

**Australia and New Zealand.** Australia has its own TPT system, HASTA, which may limit the range of products tested. However, certified supplements are generally available, with additional products often imported from Europe and the United States. To improve athlete access, an online logistics service has been established that sells only batch-tested supplements (HASTA, n.d.).

In New Zealand, most TPT supplements are imported from Australia, Europe, or the United States. However, access is occasionally limited due to global disruptions such as port congestion, rising costs, and shipping delays—particularly for European imports affected by the Red Sea conflict and capacity constraints along alternative routes such as the Panama Canal. Solutions are limited to applying a food-first strategy and/or educating the athletes to purchase TPT supplements while overseas (Europe/United States) for international competition.

**Europe and North America.** In these regions, access to TPT supplements is generally not a limiting factor. Instead, differences in supplement use tend to reflect the athlete's level. Elite athletes are more likely to use TPT-certified products, while amateur athletes rely less frequently on TPT certifications (McDaid et al., 2023, 2024; Wardenaar et al., 2020).



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**Latin America.** Certified, sealed TPT supplements are available in Latin America to some extent, but access remains significantly more limited than in Western regions. One possible reason is that regulatory agencies, such as Agência Nacional de Vigilância Sanitária, focus primarily on the internal market, making it difficult for foreign products to enter. Because fewer TPT supplements are accessible, sport associations and domestic teams often shift more responsibility onto manufacturers, requiring them to provide evidence of contamination prevention measures. To support traceability, a container from each purchased supplement is typically saved for at least 6 months in case batch testing is needed. As a result, flavor options for some products may be limited. Generally, athletes are encouraged to obtain supplements from the team nutritionists, rather than buying them themselves. However, when athletes are provided supplements from sponsors or when they purchase supplements on their own, they are encouraged to keep a sealed container for testing if needed. In Colombia, one initiative (known as the medical team practice Vademecum) involves a handbook shared with medical staff and athletes to identify substances—often referred to as "supplements"—that require stricter quality control, such as steroids, diuretics, and stimulants. To further protect athletes, medical staff and performance nutritionists also maintain detailed records of both purchased and recommended supplements.

### First-Hand Experiences and Perceptions

Misconceptions about low-risk supplement use remain common, including confusion around the responsibility of supplement manufacturers to protect athletes from anti-doping rule violations resulting from contaminated products. In regions with limited access to TPT supplements, some governing bodies have responded by advising athletes to avoid supplements altogether.

Education and awareness also vary across regions. While athletes in Western countries may be more aware of the risks of supplement contamination and inadvertent doping, they also tend to have more tools and institutional support to help mitigate these risks. Many professional teams in these regions provide low-risk supplements to athletes as a risk management strategy. At the same time, the studies on TPT supplement use have only been conducted in these countries, and it is unclear how these data can be generalized beyond Western contexts (Dow et al., 2025; Gimbar et al., 2025; McDaid et al., 2023; Schott, Seltzer, et al., 2024; Vento & Wardenaar, 2020; Wardenaar et al., 2020; Wardenaar, Lybbert et al., 2024; Wardenaar, Schott et al., 2024; Wardenaar & Hoogervorst, 2022).

Due to these limitations, oversight and responsibility are not consistently upheld across regions or organizational levels. Inadvertent doping remains a significant risk when athletes use untested nutritional supplements (Barker et al., 2025; Eichner et al., 2019; Lauritzen, 2022). Although athletes are ultimately responsible for their own supplement choices, teams and sports organizations often fall short in providing low-risk alternatives, potentially compromising athlete safety. Table 6 presents a selection of firsthand experiences and perceptions from athletes and coaches, along with consensus commentary from the authors.

## Implications and Recommendations

### Supplement Behavior of High-Level Athletes

Globally, most elite-level athletes report using multiple nutritional supplements. These most commonly include protein powders, vitamins, minerals, carbohydrate-containing sports drinks, creatine, and amino acids. Of particular concern was the frequent use of supplements considered to be "risky," due to their higher potential for containing prohibited substances. These products, which may include certain preworkouts, fat burners, weight loss products, and proprietary blends, pose a significant risk of resulting in an anti-doping rule violation to athletes, especially in elite contexts where anti-doping regulations are stringent (Jagim et al., 2023; Lauritzen, 2022; White, 2022). The inadvertent consumption of contaminated supplements has been linked to numerous anti-doping violations (Baylis et al., 2001; Kozhuharov et al., 2022; Lauritzen, 2022; Martínez-Sanz et al., 2017; Maughan, 2005), underscoring the importance of informed and cautious supplement use. Consistent TPT use ranges from 18% to 85%, based on studies conducted primarily in Western countries. Usage in other regions is likely lower, largely due to reduced availability of TPT-certified products. Another contributing factor to the low adherence to TPT-certified products could be the limited knowledge athletes have about TPT. Many athletes, and sometimes even support staff, are unaware of which certifications are reputable or how to identify them on product labels.

Additionally, the importance of improving awareness of TPT and their subsequent use was highlighted by data from the Brazilian Doping Control Laboratory that found an alarming number of contaminated samples in commercially available nutritional supplements in Brazil (Torres et al., 2024). Additionally, compound pharmacies that provide customized medications that aren't commercially available, tailoring them to individual patient needs or preferences, are commonplace in Brazil. However, these have also been shown to include contaminated supplements (Torres et al., 2024). While not fully known, this is a concept of global concern.

Current research on supplementation use and behaviors among athletes lacks methodological standardization. Existing studies employ a wide range of assessment approaches, including interviews, open-ended questionnaires, and predefined item lists. Moreover, the temporal scope of assessment varies considerably—from lifetime use to current or recent use—making cross-study comparisons highly challenging. Furthermore, although the International Olympic Committee consensus statement provides a definition of supplements (Maughan, Burke, et al., 2018), there remains no universally accepted classification system delineating what constitutes a supplement across the scientific literature. To advance this field, there is a need for the development of best-practice guidelines outlining standardized methods for assessing athlete supplementation use and behaviors, as well as uniform reporting frameworks to enhance comparability and reproducibility across studies.

### Authorities and Legislation

Authorities typically included government bodies such as ministries, national food safety agencies, and organizations responsible for supplement import and export. The legislative examples addressed food law regulations related to the manufacturing, labeling, claims, adverse event reporting and monitoring, and consumer access to safety information for (low-risk) supplements.

Across all regions, national authorities oversee nutritional supplements, and most countries apply general food safety laws in the absence of specific nutritional supplement legislation. A majority of countries have adopted (the World Health Organization) GMP standards, but they are not necessarily applied to manufacturing nutritional supplements. In many countries, the lines between food, supplements, and medicines are blurred. Overall,



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**Table 6 Firsthand Experiences and Perceptions From Regions (Anonymous) and Author Consensus Commentary**

| Experiences and perceptions | Author consensus commentary |
|---|---|
| "After a coach completed a session raising awareness about doping and presenting all the possible and legal alternatives, the coach claimed that whatever we offer as alternatives, in body building only cheaters win competitions." (Africa) | Performance nutritionists play a critical role in safeguarding their athletes' health. Although differences in nutritional supplement policy between national and international sports organizations exist, many national sports dietetic associations adhere to IOC guidelines (Maughan, Burke, et al., 2018) and WADA policy (World Anti-Doping Agency, 2021). In addition, team staff, and coaches in particular, have a significant role in building a culture of clean sport (Barnes et al., 2022). |
| "One of the recommendations of a seminar about nutritional supplements for athletes organized by the local anti-doping agency was to prohibit the use of all sports supplements and invite athletes to have a balanced diet instead." (Africa) | A food-first approach in many cases reduces the risk for inadvertent doping, although meat, poultry, and fish can contain banned substances in certain countries, including hormones and clenbuterol (Walpurgis et al., 2020). At the same time, recently more support has been gained for a food-first, but not always food-only, approach (Close et al., 2022), which includes evidence-based TPT nutritional supplement use, underlying education, and risk communication. |
| "This supplement is GMP (Good Manufacturing Practice) certified, so it is clearly free from WADA banned substances." (Australia and New Zealand)<br>"Athletes often trust supplements purchased from reputable hospitals or pharmacies, assuming they are free from banned substances. But when one athlete tested positive, it turned out that even the multivitamin supplement was not batch-tested." (Asia) | Only third-party organizations that test for banned substances can provide greater certainty that a supplement is low risk. However, testing methods and the number of substances screened can vary widely between organizations (Eichner et al., 2019). |
| "Since the 90s there is a widespread misunderstanding of the risks associated with contaminated nutritional supplements, as inadvertent doping can have profound consequences on an athlete's career and reputation, despite the absence of intentional wrongdoing." (Australia and New Zealand) | On average, ~9% of doping cases in the United States have been associated with nutritional supplement use (Outram & Stewart, 2015). Inadvertent doping through nutritional supplement use is not uncommon (Outram & Stewart, 2015). Education about evidence-based supplements and knowing where to find and order TPT supplements suitable for athletes play an important role in risk mitigation (Dow et al., 2025). |
| "Herbs and botanicals are safe, because they are natural products." (Australia and New Zealand) | Certain unregulated supplements—including some herbs and botanicals—can pose serious health risks, even when marketed as safe. In some cases, their use resulted in hospitalization and surgery (Hassen et al., 2022). |
| "Because our nutrition sponsor is well-known in the world of cycling and expert in food safety, there is no need to test every single product and worry about labels." (Europe)<br>"There are teams that work with specific brands that they trust, mainly due to having used them previously and not having returned positive doping tests." (Latin America) | Strict liability for a positive doping test applies to athletes competing under WADA regulations (World Anti-Doping Agency, 2021). As such, sport teams and organizations should maximize their effort to ensure the nutritional supplements provided are properly tested for banned substances (Maughan, Burke, et al., 2018). |
| "Supplement contamination is a top concern as a nutritionist of the women's national team and a top domestic women's football team, but access to tested supplements is limited. Therefore, we try to buy supplements in bulk when available, this means that we sometimes run out and don't have a suitable alternative." (Latin America) | Global access to TPT supplements suitable for athletes is limited, making systematic nutrition education and clear guidance on athlete responsibility even more critical. In elite sport, only tested supplements should be used—even if that means going without. While athletes should prioritize optimal nutrition, a food-first approach may not always meet needs for certain compounds, such as beta-alanine or creatine. |
| "Although athletes are advised not to obtain individualized supplements, many visit compound pharmacies that put together a mix of substances." (Latin America) | Pharmacies lack anti-doping oversight, there is a risk of cross-contamination with products not suitable for athletes, and supplements are not third-party tested for banned substances (Torres et al., 2024). Athletes are strongly discouraged from sourcing tailor-made supplements from pharmacies (Greenbaum et al., 2023). |
| "A concern of our medical team is that athletes think that all products are equal in quality or safety." (Latin America) | Nutritional supplements are not equally safe, as at least one out of five supplements contains products not listed on the label (Duiven et al., 2021). Proper athlete education and a policy on the team level on how to select and source low-risk nutritional supplements are recommended. |
| "National team athletes belong to international clubs, where they not only have a broader range of supplements, but each team also has a different periodization for using the products, adding another layer of complexity in safeguarding the athletes supplement portfolio." (Latin America) | When athletes are part of different teams or programs, performance nutritionists need to rely more on the responsibility of the athlete. However, performance nutritionists should connect across teams to discuss supplement policy and always employ safe and low-risk supplement practices, including the use of TPT supplements. |

*(continued)*



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**Table 6** ***(continued)***

| Experiences and perceptions | Author consensus commentary |
|---|---|
| "With the good intention of improving their children's performance, it is worrying to see how some parents of competitive athletes buy supplements that lack the necessary safeguards and become involved in doping cases." (Latin America)<br>"I found supplements in the kitchen and my parents use them. The container stated that they promote general health. So, I assumed it will be good for me." (Europe) | Athletes are easily influenced by their social network (Wardenaar, Lybbert et al., 2024), specifically at a younger age when parents or caretakers are responsible for purchasing nutritional supplements (Wardenaar et al., 2024). Therefore, this group should be included in safe and low-risk supplement education (Dow et al., 2025). |
| "There are so many different quality certificates that it is difficult for athletes to understand if and how they are relevant to keep them safe from inadvertent doping." (North America) | Sports teams and organizations should educate athletes on how to select the right TPT organizations (Eichner et al., 2019) and the difference between testing for WADA Prohibited Substances and other testing. |

*Note*. TPT = International Olympic Committee; WADA = World Anti-Doping Agency.

regional harmonization differs. Europe leads with unified laws (excluding botanicals), while Asia, Africa, and Latin America show fragmented approaches. In North America, despite the lack of regulatory harmonization, both the United States and Canada have well-developed systems for overseeing nutritional supplements and ensuring product safety. Australia stands out for clearly classifying supplements as either food or therapeutic goods, whereas Africa lags with outdated or minimal regulations.

## Systems Available

There is alignment and contrast between regional approaches to supplement safety and education. In Europe, North America, and Australia, there are robust testing infrastructures with TPT-certified products widely available. While regions such as Africa, Asia, and Latin America rely more on certification services from Western regions, access is limited by cost and availability. Education activity is present in all regions, but its reach and integration with anti-doping efforts vary. Europe and North America offer deeply institutionalized programs, whereas Africa and Latin America focus more on outreach and regional networks. Despite these differences, all regions show a shared commitment to improving athlete awareness and reducing doping risks, and NOCs and NADOs are the common denominator spreading these messages.

Certified supplements may cost more, specifically in low-income countries, and combined with sponsorship disparity, access to TPT supplements is often restricted. Regulatory approaches vary across countries, and language barriers may further limit access to appropriate supplements for some populations. Importing TPT supplements can be costly, slow, or legally restricted. Finally, and as a result of the aforementioned barriers, counterfeit products labeled as tested could appear on the market.

## First-Hand Experiences and Perceptions

Despite the availability of strong educational resources, anecdotal insights reveal widespread misconceptions about low-risk supplement use and the extent to which athletes can prevent positive doping tests. In regions with limited access to TPT products, some sports organizations recommend avoiding supplements altogether. Awareness and education around inadvertent doping also vary by region. While athletes in Western countries may have a better understanding of contamination risks, this likely reflects the fact that most research on safe and low-risk supplement use has been conducted in those regions. Although athletes are held individually responsible under strict liability policies, teams and organizations often fail to provide adequate support or access to low-risk supplements—ultimately putting athletes at risk.

## Strengths and Limitations

A clear strength of this article is that it provides a comprehensive global overview of nutritional supplement use and safety in (high-level) athletes, demonstrating the diversity of regulatory frameworks and the common challenges encountered across six defined regions. Furthermore, the engagement of experts coming from each of these regions adds important contextual detail, particularly in areas lacking unified legislative structures. Finally, by visualizing the empirical findings alongside field-based anecdotes, this article offers a nuanced picture of practical concerns, such as nutritional supplement contamination and inconsistent education, that affect athletes worldwide. Overall, the outcomes provide valuable insight for future policy development and organizational planning.

However, the article also has its limitations. Due to the lack of regional harmonization, the actual regulation of nutritional supplements often takes place on a national level. The reliance on regional representatives, rather than a systematic analysis of individual countries, may overlook specific national contexts or recent policy shifts. Additionally, the data underpinning supplement use and doping risks are more robust in Western regions, which may lead to geographical bias in the findings. While the firsthand experiences and perceptions enrich the narrative, their subjective nature limits generalizability, and the list is by no means complete. The article's broad scope, though a strength in coverage, inevitably constrains detailed examination of regional legal, cultural, and infrastructural complexities.

## Takeaways for the Future

While it is widely agreed that athletes should prioritize balanced, nutrient-rich diets, a "food-first, but not food-only" approach is increasingly accepted (Close et al., 2022). In certain cases, the use of nutritional supplements may be appropriate, provided that their safety and necessity are carefully considered (Maughan, Shirreffs, et al., 2018). Global findings reveal that access to TPT supplements and education around supplement safety are uneven, with athletes in lower-resourced regions facing greater risks. While some countries benefit from strong regulatory oversight and well-established educational systems, others lack the infrastructure to ensure safe and low-risk supplement use. These disparities underline the need for collective, system-wide efforts to protect athlete health and integrity.

To Move Forward, Several Actions are Essential:

1. A Pragmatic Dietary Approach: Athletes should be encouraged to prioritize whole foods, but the responsible use of supplements, when needed, should be supported with proper guidance as well as safety and risk checks.



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2. TPT for Banned Substances is Critical: In the absence of access to major commercial testing bodies, certification through independent, accredited laboratories conducting analysis for WADA Prohibited Substances is essential for any athlete subject to doping controls. Furthermore, standardization of the number of screened WADA prohibited substances, batches tested, and/or GMP audits for TPT organizations would be of value.
3. Shared Responsibility: Although athletes are ultimately responsible for what they consume, low-risk supplement use must be a collective effort. Coaches, nutritionists, and sports organizations, as well as parents of youth athletes, must contribute to creating environments that support informed and low-risk decisions.
4. Education is Fundamental: Misinformation and knowledge gaps persist across all levels, including athletes, parents, and support teams. Structured education programs for all age groups, aligned with anti-doping guidelines, are necessary to close these gaps.
5. Behaviorally Informed Strategies: To effectively reduce the risk of inadvertent doping, interventions must go beyond individual-level education. A systems-based approach can help map the complex interactions between athletes, support teams, commercial pressures, and regulatory gaps. In parallel, behaviorally informed strategies—such as nudges, decision aids, and environmental cues, including the provision of low-risk supplements—can support safer athlete choices in real-world settings.
6. Alignment and compatibility among third-party certification programs that test for banned substances are currently limited. Greater standardization across organizations is needed to ensure consistency in certification quality while minimizing disparities in accessibility of certified products. For example, some certification programs offer user-friendly tools, such as mobile applications that allow athletes to scan a product's barcode to verify whether a specific batch has been tested. In contrast, other programs require users to manually enter batch numbers on a website, creating additional barriers. These inconsistencies may discourage athletes from verifying supplement certification status and, consequently, reduce the likelihood of selecting products that are less likely to be contaminated.

## Conclusion

The findings of this region-based focused overview suggest the importance of a broad global acknowledgment of the importance of regulating and educating athletes on the use of nutritional supplements. However, the extent and effectiveness of these efforts differ significantly across regions. While regulatory frameworks focusing on manufacturing low-risk nutritional supplements are present worldwide, only certain areas—such as Europe, North America, and Australia/New Zealand—demonstrate the consistent availability of TPT systems. In these regions, additional efforts are made to facilitate decision-making toward the use of effective supplements, as well as efforts to influence nutritional supplement behavior. In contrast, athletes in other regions often face limited access to reliable TPT and structured educational programs, increasing their exposure to inadvertent doping risks. These findings underscore that ensuring supplement safety is not merely a matter of individual responsibility. By framing supplement safety as a shared responsibility and building stronger systems for education and access, the risk of inadvertent doping can be significantly reduced.

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## Nontechnical Summary

Nutritional supplements are widely used by athletes around the world to support performance, recovery, and health. Commonly used products include protein powders, vitamins and minerals, creatine, caffeine, and carbohydrate-based sports drinks. While many of these can be safe and effective when used appropriately, supplements can also carry risks—especially when products are contaminated with substances banned in sport. Such contamination can lead to unintentional anti-doping rule violations, with potentially serious consequences for athletes' careers.

This paper provides a global overview of how athletes use supplements, what types of regulatory systems are in place to ensure their safety, and how programs that independently test supplements—known as third-party testing (TPT)—can help reduce the risk of contamination. The perspective was written up coming from six major world regions: Africa, Asia, Australia/New Zealand, Europe, Latin America, and North America. Information came from scientific studies, official government documents, and expert contributions from each region.

Across these regions, supplement use among athletes varies widely—from as low as 7% to nearly 100%—depending on sport, level, and local factors. Basic supplements such as protein powders and multivitamins are commonly used globally, but also higher-risk products such as preworkout formulas, herbal blends, weight-loss products, and proprietary blends are frequently reported by high-level athletes but to a lesser extent. These types of products are of concern because the literature has suggested that they are more likely to be contaminated with banned substances or include ingredients not listed on the label. Unintentional doping violations can result from taking such supplements, highlighting the need for better education and safer options.

Many countries have systems in place to regulate supplements, but the strength and enforcement of these systems differ greatly. In Europe, North America, and Australia/New Zealand, there are well-established frameworks to monitor supplement manufacturing and labeling. These regions also have broader access to TPT programs that verify products for prohibited substances. In contrast, Africa, Asia, and Latin America often have less comprehensive regulations and limited access to certified, low-risk supplements. Cost, lack of local testing infrastructure, and language barriers further restrict access to safe products.

Third-party testing programs play a crucial role in reducing inadvertent doping risk as a result of supplement use. The programs recommended for athletes test nutritional supplements independently to ensure that they are free from banned substances. Products that pass these tests receive certification marks, helping athletes make safer choices. However, these programs are unevenly distributed worldwide and differ in how they operate. In some cases, athletes can easily check if a product batch is certified by scanning a barcode, while in others, verification requires searching online databases using batch numbers. This lack of consistency can discourage athletes from checking certifications and reduce the impact of these programs.

Despite increased awareness in some regions, many athletes and even members of their support teams—such as coaches or nutritionists—remain unaware of which certification programs are trustworthy or how to recognize them. Misunderstandings about supplement safety are common, and in some countries, limited access to certified products has led sports organizations to recommend avoiding supplements altogether. However, this approach can also limit athletes' ability to use evidence-based supplements safely and effectively.

Authorities responsible for supplement oversight typically include national food safety agencies or ministries of health. In many countries, supplements are treated under general food laws rather than having their own specific legislation. Although many nations claim to follow international Good Manufacturing Practice (GMP) standards, these are not always applied consistently, especially in lower-resourced settings. Some regions, such as Europe, have developed more harmonized approaches, while others—particularly in Africa and Latin America—still face fragmented or outdated systems.

Across all regions, educational efforts about supplement safety exist but vary greatly in reach and quality. In Europe and North America, supplement education is often integrated into anti-doping and sports nutrition programs. In contrast, other regions rely more on regional workshops or outreach efforts that may not reach all athletes. Despite these differences, there is a shared global commitment to improving athlete education and reducing doping risk.

Importantly, supplement safety should not be viewed as solely the athlete's responsibility. While anti-doping rules hold athletes accountable for what they consume, ensuring access to low-risk supplements requires cooperation among athletes, coaches, sports organizations, nutrition professionals, and regulators. A shared responsibility approach would better protect athletes from accidental doping.

To move forward, several actions are needed:

1. Encourage a "food-first, but not food-only" approach: Athletes should prioritize balanced diets, but when supplements are necessary, they should be used responsibly with guidance from qualified professionals.
2. Expand and harmonize TPT: Broader global access to independently tested products is essential, particularly in under-resourced regions. Standardizing testing criteria—such as which banned substances are screened and how often batches are tested—would strengthen trust in these systems.
3. Promote education for all stakeholders: Athletes, parents, coaches, and support staff should receive structured education on supplement safety and how to identify certified products.
4. Address behavioral and systemic factors: Beyond education, practical strategies—such as simplifying certification checks and making low-risk products easily available—can help athletes make safer choices.
5. Enhance policy and collaboration: Governments, sports organizations, and industry partners should work toward harmonized safety standards and improved regulation of supplement manufacturing and marketing.

In conclusion, this global overview highlights that nutritional supplement use among athletes is widespread but unevenly supported by safety systems and education. While many athletes can access well-regulated products and reliable TPT, others—particularly in lower-resourced regions—face greater risks due to limited oversight and availability of certified supplements. Ensuring safe and informed supplement use requires shared responsibility among athletes, support staff, organizations, and regulators. Strengthening education, expanding access to tested products, and aligning global safety standards will be key to protecting athlete health and integrity while supporting fair competition worldwide.



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